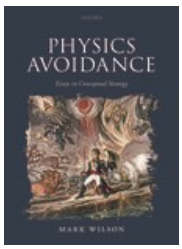

Physics Avoidance: and other essays in conceptual strategy

Mark Wilson

https://doi.org/10.1093/oso/9780198803478.001.0001

Published: 02 November 2017 **Online ISBN:** 9780191841651 **Print ISBN:** 9780198803478

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CHAPTER

8 Semantic Mimicry

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https://doi.org/10.1093/oso/9780198803478.003.0008 Pages 324–361

Published: 02 November 2017

Abstract

Reasoning patterns can look superficially alike, yet differ significantly in their supportive underpinnings. Paradoxically, the progress of science is sometimes aided by these inferential pretenders, in spite of the conceptual confusions upon which they rely. Two puzzling developments of this character are studied from this perspective: the "little plates" of the aircraft engineer and the incoherent infinitesimals of Physics 101 instruction. The proper semantical foundations of both notions weren't unraveled satisfactorily until the mid-twentieth century, when the challenges posed by the finite element revolution within computing were squarely addressed by mathematicians. Their results teach a valuable lesson in how our classificatory vocabularies sometimes divide descriptive duties amongst themselves in very intriguing patterns of cooperative allocation.

Keywords: physical infinitesimal, fundamental solution, finite element, distributions, functional analysis, delta function

Subject: Philosophy of Science, Philosophy of Language

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> *The interesting point here is that whilst many problems of applied mathematics are posed in terms of differential equations, the finite element solutions of such equations utilize ideas which are in fact much older than those used to set up the equations initially.*
>
> A. J. Davies[1]

(i)

When I was a teenager, my brother George would bring home philosophy books from college, including W. V. O. Quine's Methods of Logic and Israel Schleffer's The Anatomy of Inquiry.[2] For reasons known only to the juvenile mind, I gobbled these up. To be sure, I suspected that Scheffler's specimen illustrations of "laws of

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nature" (e.g. "phosphorous smells like garlic") probably did not represent the stuff of which science is truly made. Given the inadequacies of my high school tutelage, I had no plausible replacements to offer. During my first week of Physics 101 in college, everything went well enough, commencing with Newton's Three Laws of Motion and all that (although that Third Law struck me as rather dodgy). But soon we encountered loaded strings and beads sliding along wires and, for the life of me, I could not figure out how the announced "theorems" followed from the "axioms" presented (I had acquired a sound grounding in first-order logic from my brother's books). I dimly recognized that ↳ some meritorious wisdom must lie latent within these "illogical" procedures, but I was damned if I could see what it was.

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George and Mark

I asked my instructor about these oddities, but he only snapped, "Science always idealizes; go do your homework." Traditional responses like this have led many generations to the despairing conclusion, "One never understands physics; one only learns to do it."[3] Only many years later did I learn that these strange but venerable derivational tactics have proved of great concern to serious practitioners of classical mechanics, prompting Stuart Antman's complaint:

> *Most modern expositions* [of mechanics still] *... ask the reader to emulate the Red Queen by believing six impossible things before breakfast.*[4]

These woes are not merely a matter of intellectual tidiness. Without a clearer understanding of how the task of articulating modeling equations for a continuous material should be approached, early twentieth-century materials scientists found themselves lacking the guidance needed to deal with more complex materials than the stereotyped successes of textbook pedagogy.

As exemplars of these "impossible things," we might consider some of the assumptions my instructor asked me to swallow setting up the equation for the equilibrium of a loaded string: $\mathrm{k}\mathrm{d}^2\mathrm{y}/\mathrm{dx}^2 = \mathrm{f}$. This formula encapsulates the claim that the string locally counterbalances an applied gravitational load f by generating an internal stress resisting its degree of curvature $\mathrm{d}^2\mathrm{y}/\mathrm{dx}^2$. But the standard derivational pathway I was taught begins—see left side of the illustration—by initially considering an extended length of string—called an "element"—and then squeezing its two endpoints closer together until they collapse into a single point possessing the required curvature $\mathrm{d}^2\mathrm{y}/\mathrm{dx}^2$. Along the way, we are exhorted to accept the following "impossible things":

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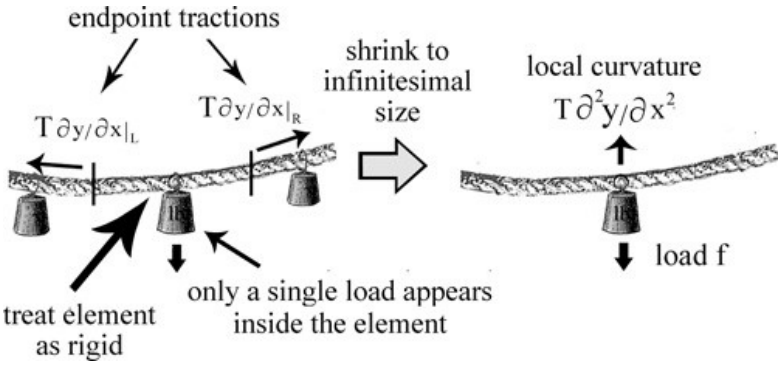


setting up the loaded string equation

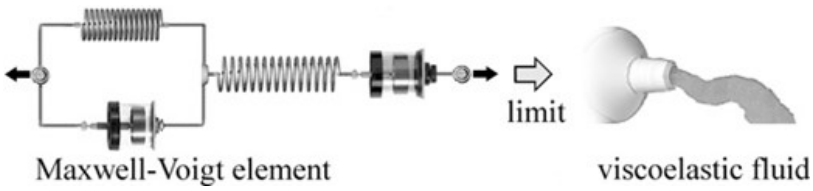
(1) *At equilibrium at small scales, a flexible piece of string can be treated as if it were a rigid body.*[5]

p. 326 (2) ↳ *When distorted, strings move entirely in a vertical direction.*

(3) *Any special events occurring at the boundaries of the element disappear when a final limit is taken.*

(4) *Our element can be chosen small enough that it only bears a single gravitational load yet its endpoints remain distinct.*

At the time I became too discouraged by my lack of understanding to advance far enough into mechanics to learn about Maxwell–Voigt elements for viscoelastic fluids, but if I had, I would have been surely perplexed by the assertion that if one only squeezes these arrangements together into a limit, an appropriate model for toothpaste will emerge!



a strange dissection into elements

Further perplexities appeared when later in the course we attempted to model a continuous material in three dimensions, such as a large hunk of steel. These considerations made me reflect back on earlier days and wonder, "Shouldn't we have started with this three-dimensional modeling and obtained its one-dimensional string equation cousin through some sort of dimension-dropping rationale? Violin E strings, after all, are made of exactly the same kind of steel."

The proper answer to my question is "yes, we should," but following such a policy would have required my instructor to replace his several lines of efficient, if mystifying, derivation of the original string equation with lengthy pages of elaborate mathematical manipulation. Methodological reappraisals of this sort are common within the history of science—what initially appeared to be a simple inferential transition from sentence A to sentence B turns out to secretly depend upon a large amount of hidden machinery, whose elaborate workings must be detailed if we are going to infer B reliably from A.

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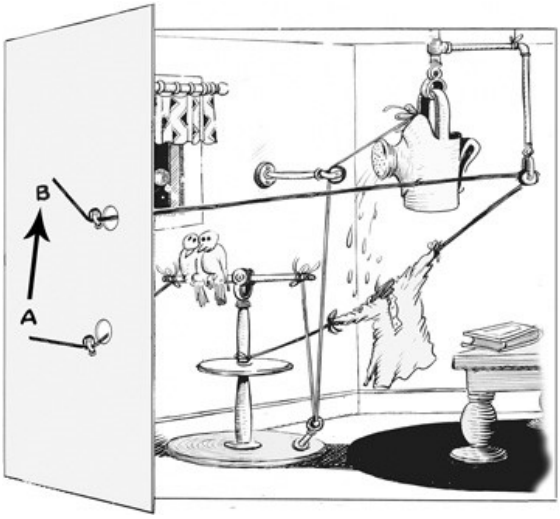

hidden machinery behind a simple inference

Such are the circumstances of *semantic mimicry*: situations where a language's employers fancy that sentences A and B signify physical underpinnings of a certain ↳ type, when the real-world support of their modeling successes lies elsewhere. Surface syntactic resemblances sometimes deceive.

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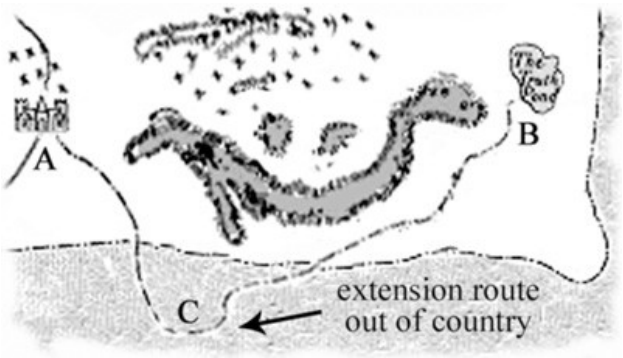
Note the term "reliably." We should recognize that various forms of derivational safeguard must have lain latent within the contextual woodwork that allowed crude forms of A to B reasoning like my instructor's to reach credible goals on a fairly trustworthy basis. After all, all the early masters of mechanics followed similar disreputable procedures. Many subsequent improvements in our understanding of applied mathematics have permitted an identification of these hidden conceptual controls. Many of these first appear as "extension elements" that allow us to improve a derivational landscape in the same manner that adding complex points and points at infinity to the regular Euclidean plane produces significantly improved deductive efficiency (e.g. by allowing us to draw "lines" to these supplementary points in the same fashion that we utilize regular positions within traditional geometry). Sometimes the most efficient deductive pathway from location A to B travels through a position C that lies outside of one's originally contemplated borders.[6]



improved reasoning via extension elements

(ii)

Here's a typical illustration of how such extension elements improve our inferential lot in life, which will later prove pertinent to my Physics 101 worries. The two-dimensional analog to our string is the loaded membrane, $k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f$. As discussed in Essay 1, we normally select the boundary conditions appropriate to

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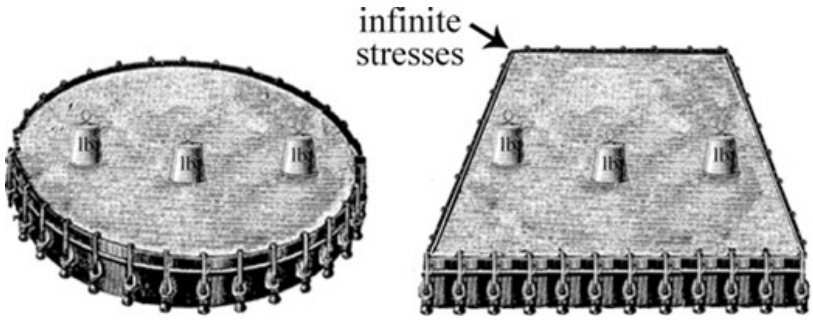
a problem based upon our sense of the natural possibilities available here—isn't it obvious that we could cut a sheet of parchment to any shape we desire and tug on its edges in any manner we find fit? However, simple choices of shapes like squares will not harmonize properly with that interior equation, understood in a traditional way.[7] Why? If one shears the edges of our square in a straightforward way, the interior equation says that infinite tensions will get created at its four corners, rendering the anticipated match-up between boundary and interior nonsensical.



loaded drumheads

p. 328 The favorite tactic of the careless methodologist is to brush such concerns aside through blithe appeals to "idealization": "Oh, nothing in nature really possesses such sharp corners; so let's just stick to boundary curves that suit our interior equations." I remember being puzzled years ago by such responses. I wondered, "The interior equation description seems to be every bit as 'idealized' with respect to its underlying physics as its boundary assignments; perhaps even more so. So why should our natural choices for boundaries get overruled by the interior?" As we'll see, such worries are entirely cogent and motivate experts such as J. N. Reddy to warn:

> *It should be understood that the equations governing a physical problem are themselves approximate. The approximations are introduced through several sources, including representation of the geometry and boundary conditions, loads and material behavior. Therefore, when one thinks of permissible error in an approximate solution, it is understood to be relative to exact solutions of the governing equations that inherently contain various approximations.*[8]

In any event, historically it was soon recognized that dismissing our perfect squares altogether in this prim fashion results in a significant loss of pertinent information, as G. Barton explains:

> *Of course, in practice the* [Green's function procedure] *is freely applied even to regions that do have corners and edges: for instance, we often deal with rectangular boundaries. While the mathematician might regard such* [boundary value problems] *as ill-defined at a corner, the physicist takes the explicit expression* [derived formally by the Green's function method] *as his solution; if he needs* [a value for a quantity] ψ *near a corner, then he simply consults the* [formal expression for ψ] *in order to see what the solution is. The justification is that, on the one hand, the corners of real physical boundaries, like metal boxes, are not mathematically sharp; while, on the other hand, only in regions negligibly small by normal standards is there any appreciable difference between the solution calculated for the true shape with rounded corners, and the idealized solution calculated for idealized rectangular shapes, albeit ignoring any mathematical ambiguities at the corners.*[9]

p. 329 I'll explain what he means in a moment. But let us observe that his "Green's function formal solutions" here operate as out-of-territory extension elements, whose historical role in keeping physical reasoning rolling along profitable rails is a topic we shall explore.

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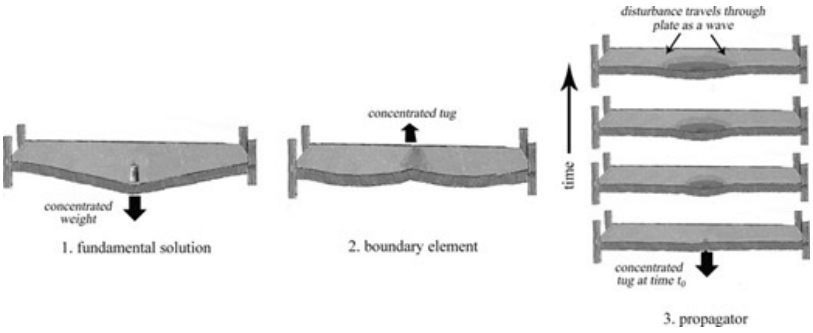
As our mathematical understanding has advanced, it has been gradually recognized that relying upon the inferential labor of these out-of-territory ingredients without proper acknowledgment is unjust—their operations should be treated as vital components within an integrated frame that also embraces the familiar formulas of officially sanctioned practice, such as the interior equation $k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f$. This shift in point of view is a matter of simple informational justice—cast off your chains of oppression, unacknowledged guest workers! Accordingly, we must seek improved pictures of how the various sentences appearing within our successful inferential practices encode pertinent information about the physical world. Often suitable answers are both sophisticated and unexpected. Formulas formerly regarded as mere "approximation statements" turn out to encode central physical considerations that allow our language to track reality ably. The purpose of this essay is to illustrate what this hidden machinery looks like when flushed out into the open air.

unrecognized labor

Let's go back to the Barton quote. What does he mean by a "Green's function"? Well, they fall into three major classes that we should properly distinguish.



three kinds of Green's functions

(1) A mathematical formula $g_\alpha(x, y)$ that reflects how a point load singularity within the *interior* of our membrane at location α will weigh the rest of it down ↳ under some prescribed boundary moorings at equilibrium. These rules are often called "influence functions" as well (I shall usually call them "fundamental solutions").

p. 330

(2) A mathematical formula that reflects how a singular tug along the *boundary* of our membrane will distort an otherwise undisturbed interior at equilibrium. Such rules are often called "boundary value elements."

(3) A mathematical formula that reflects how a singular disturbance applied within the interior of an

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otherwise undisturbed membrane at time t_0 will *evolve over time* (e.g. traveling as a wave). Such rules are often called "propagators." Note that, unlike (1) and (2), we are no longer restricting our attention to equilibrium. Indeed, no measure of passing time enters the circumstances of (1) or (2) at all.

In the quotation cited, Barton is talking about Green's functions of type (2), but we'll initially confine our attention to formulas of type (1), as they are easier to handle. But the basic observations apply, mutatis mutandis, in all three cases. By traditional standards, our so-called "fundamental solution" $g_\alpha(x, y)$ can't qualify as a true solution at all, because $g_\alpha(x, y)$ can't obey the interior equation at the loading point α under any classical understanding of "obey." Why? Because our membrane equation demands second partial derivatives ($\partial^2 z/\partial x^2$ and $\partial^2 z/\partial y^2$) everywhere in the interior, but $g_\alpha(x,y)$ doesn't even possess first derivatives at α, due to the concentrated loading. Nonetheless, with $g_\alpha(x,y)$'s assistance, we can subsequently reason our way to solutions for tamer forms of loading h(x,y), such as a load distributed over a finite stretch of membrane. We do this by simply multiplying the $g_\alpha(x,y)$ and h(x,y) together at every point and summing up the results.[10] This multiplication by h(x,y) weakens the strength of the $g_\alpha(x,y)$ singularity enough that the resulting formula can genuinely satisfy our membrane equation, under appropriate restrictions on h. In such circumstances, $g_\alpha(x,y)$ is said to act as a "kernel" inside the integral. Essentially, the same trick leads to good results for finite boundary tugs if we employ a suitable Green's function of class (2).[11] We obtain a good deal of valuable information about our plate from these kernel solutions.

Well, why can't we do the same thing employing some rounded off surrogate g* to $g_\alpha(x,y)$? The answer is because such g*'s rarely possess convenient analytic formulas at all and can only be computed by strictly numerical methods. As such, we have not been supplied with a g* that can serve as a kernel in the fashion just discussed. If we throw out our Green's fundamental solution formula on the grounds that it doesn't qualify as a "true solution," we deprive ourselves of an inferential pathway that efficiently leads to very important
p. 331 conclusions about our target system. Mathematicians call the algebraic ↳ information contained within a Green's function formula "analytic data," whose value they contrast with the less informative "numerical data" one obtains from a conventional computer simulation. In terms of our extension element metaphor, our algebraic $g_\alpha(x,y)$ expression serves as a key out-of-territory way station along a very profitable inferential pathway.

What went wrong in our original appraisal of our membrane's modeling? A modern diagnosis holds that we mistakenly viewed the "semantical contents" of the sentences appearing in our reasonings as individually interpreted claims, rather than as a collection of statements that must work together within a balanced descriptive network. In particular, we originally regarded our original formulation of how the *interior* of our membrane behaves (recorded in the differential equation $k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f$) and our original formulation of how its boundary region behaves (as a geometrical curve loaded with tugging forces) as completely independent descriptive tasks. But this is wrong; our naïve assumptions with respect to the "meanings" of these formulas were misguided and must be reconsidered from a more integrated and functionalist point of view. In the technical literature, this revisionist methodology is articulated in terms such as these:

> *When an engineer deals with some mechanical structure, he carries a mental image of how the structure deforms or behaves dynamically under a certain range of loads. Rather than regarding the solution to his boundary value problem as a collection of pointwise calculated values, he concerns himself with how the solution* as a whole object *changes under certain changes in the loads. This is very similar to how one considers a mechanical problem by the methods of functional analysis. In this approach a load is defined pointwise but is considered as a whole object—an element of a functional space. Next, a unique characterization of the structure's response to this load is identified; suppose it is the displacement field u*

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> *corresponding to the load F. While u is a function (or vector function) depending on the space coordinates and possibly on time, it can also be regarded as a whole object—another element of some functional space. The relation between F and u can be written quite simply as u= u(F), provided that it is remembered that this is not a pointwise dependence: here, u in total depends upon F as the entire set of loads acting on the structure.*[12]

Our task in the rest of this essay is to explicate the recommendations invoked here.

At root, our misunderstandings of semantic content trace to an underlying doctrine that we might dub *single sentence idiolatry*: the assumption that the descriptive formulas we employ in science register physical information straightforwardly on an independently established sentence-by-sentence basis, rather than working together as an integrated descriptive team. We are familiar with the comic motif—Carl Barks' *The Sheriff of Bullet Valley* can serve as an evocative exemplar—in which some doofus of a pater familias (or, in Barks' case, an avunculus familias) fancies that he is completely responsible for all of the good works achieved
p. 332 within his ambit. In reality, an unnoticed gaggle ↳ of helpful nephews secretly scurry about behind the scenes, rectifying his blunders through sage consultation of the Junior Woodchuck's guidebook to subtle strategizing. Just so: this is exactly how the modern mathematician looks upon the pretensions of modeling equations like $\mathrm{k}(\partial^2\mathrm{z}/\partial\mathrm{x}^2 + \partial^2\mathrm{z}/\partial\mathrm{y}^2) = \mathrm{f})$ regarded as stand-alone descriptive entities. "I completely encapsulate all of the physics relevant to a membrane and am wholly responsible for all explanatory successes achieved within my ambit," our formula brags. "No, you conceited fool," the mathematicians object, "you operate with your attached side conditions as a team and if our present understanding of your 'descriptive content' can't render their semantic contributions proper credit, then our appraisal of your 'content' must be readjusted until a better harmony is reached." As this improved understanding of our descriptive practices is obtained, out-of-territory reasoning ingredients formerly tolerated as "useful idealizations" emerge as integral parts of a smoothly functioning portion of descriptive machinery.[13]

(iii)

Before we look into the hidden semantics behind those puzzling derivations of my Physics 101 class, let us add a revealing historical episode to our congeries of consideration. As briefly noted in Essay 1, bridges built of rigid girders in the top manner illustrated (a so-called determinate bridge) supply engineers with a splendid computational opportunity—the basic forces and torques acting on the structure's joints can be calculated as soon as the loads upon the bridge (e.g. a locomotive) have been assigned. Once these endpoint forces and torques are known, engineers can calculate the stresses inside each member by relying upon the simple "engineering beam" rules found in every elementary text on structural analysis. These are simple algebraic
p. 333 rules that approximate the results of what happens inside a ↳ member when its endpoints are compressed or stretched. As they do this, they reason entirely in a calculate-the-nodal-values-first-and-worry-about-their-interior-parts-later manner. When they turn to the second bridge illustrated (a so-called indeterminate structure), they can utilize the same modeling ingredients but must now solve a large collection of algebraic equations simultaneously, a lengthy task best accomplished on a computer.[14]

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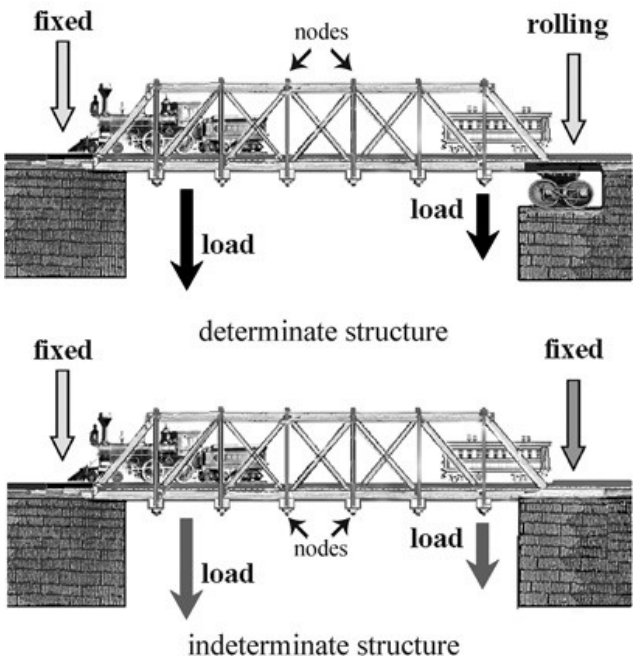
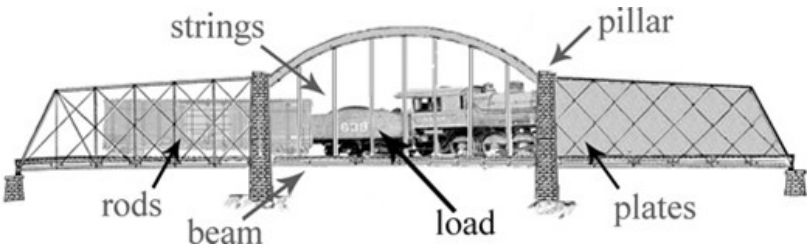


standard structural analysis

These techniques (usually called "structural analysis") were developed in the nineteenth century by Castigliano, Maxwell, and others, to a high degree of effectiveness. Central to these methods is the assumption that its target systems are entirely comprised of one-dimensional stringers joined at their nodes. Accordingly, such methods do not automatically extend to covered bridges incorporating two-dimensional plates as illustrated on the right of the diagram. This limitation naturally frustrated designers, and in the 1940s various bold aircraft engineers began to imitate standard structural analysis policies by decomposing a complicated shape like an airplane wing into little plates[15] that interact with one another along their edges. Once again they attempted to compute stresses inside each little square from its edge behavior by employing an engineer's algebraic approximation to a standard elastic plate model. In doing so, they obtained tolerable, if not perfect, estimations of stress distribution in their airplane wings. J. T. Oden writes:



will such techniques work in two dimensions?

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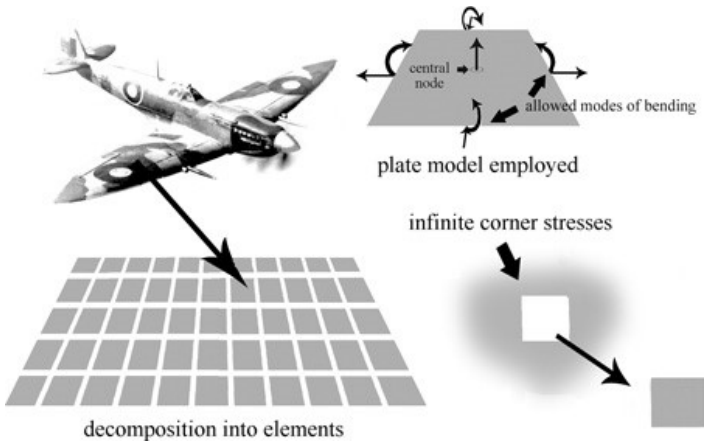
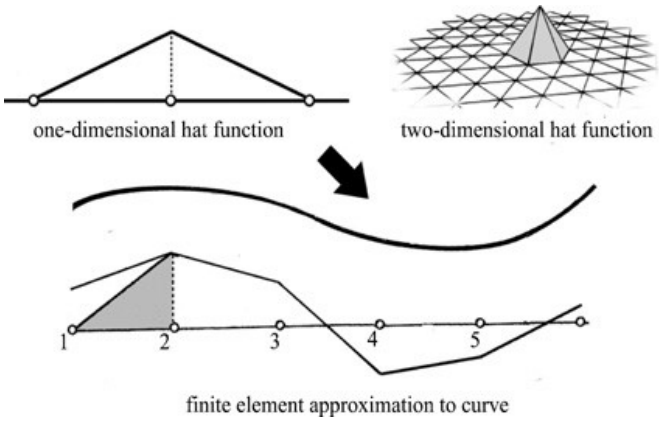


decomposition into squares of an airplane wing

> *The practice of representing a structural system by a collection of discrete elements dates back to the early days of aircraft structural analysis, when wings and fuselages, for example, were treated as assemblages of stringers, skins and shear panels.*[16]

p. 334 The basic idea was this. Let's concentrate upon the dominant manner in which adjacent plates affect one another, in particular to the manner in which tugs and twists referred to in the middle of each side will raise or lower the center of the affected square (theses specific regions are called nodes). But how did they obtain a suitable rule for raising and lowering the central node? By simplifying the boundary element Green's function for a square so that it becomes a simple algebraic relationship. In so doing, our engineers believed they were merely imitating the manner in which the structural engineers had learned to replace more sophisticated beam models with a simple "engineer's beam formula" when they analyzed a bridge in the Maxwell/Castigliano fashion.

We might observe that within modern practice it is more common to tesselate our target structure wing into little triangles and study how neighboring stresses will raise or lower a little "hat function" spanning these triangles. But to explain the rationale behind this shift now would be to get ahead of my story.



modern triangular elements

In a one-dimensional finite element setting, these hat functions turn into simple raised or lowered triangles as shown. When these rules operate conjointly we obtain broken line approximations to curves as illustrated.

In truth, my instructor had tacitly appealed to a similar decomposition in Physics 101 when he discussed our string: he effectively broke it into little elements and asked how gravity and its endpoint tensions would affect

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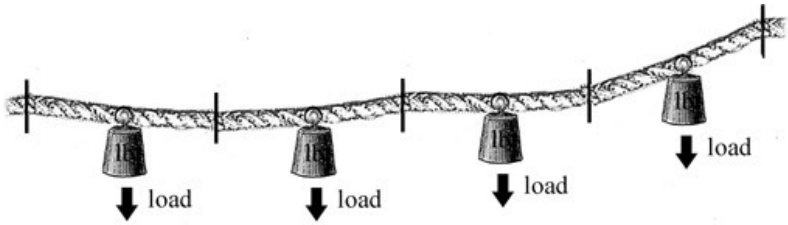
their interior nodes. At the time, this procedure struck me as weird given that our string is flexible, unlike a wire which remains largely rigid like our bridges. He would have probably mumbled something to the effect that at equilibrium flexible bodies can be modeled as if they were rigid, a strange tenet with a venerable pedigree.[17] We'll later trace a deep linkage between these veins of classical physics thinking and finite element technique.



Physics 101 finite element

Such were the stories our aircraft engineers told themselves. But something was surely wrong. If we cut a square hole in an airplane wing, its sharp corners will concentrate applied stresses significantly—indeed, the problem's non-rigorous ↳ "solution" claims that the corner stresses become genuinely infinite. In real life, these high-stress concentrations prove a significant problem: the early Liberty ships of World War II cracked apart because poor planning had outfitted them with square portholes. So the real-life physics of how a square plate welded to a square hole will behave is a very complicated affair, whereas our 1940s optimists proceeded on the assumption that dividing their airplane wings into squarish elements would create no significant effects whatsoever. In effect, they were acting upon the same presumption that my Physics 101 instructor made when he assumed that introducing boundary joins between the elements of a previously unsegmented string would make no difference to its behavior—as if pearl necklaces and strings behave alike. Of course, designers of the time were aware of the corner stress worries but brushed them aside with airy appeals to "Saint-Venant's principle" and the like: "Oh, as long as we get far enough away from those problematic edges, the platelet interiors will respond to their neighbors in an averaged manner that we can capture in suitable algebraic relationships" (Barton invokes a similar appeal in the quotation above). Resting the reliability of aircraft design upon such dubious appeals seems a rash policy but, by and large, our engineers reached creditable numerical values through such tactics in lieu of any alternative method for reaching comparable results.

p. 335

Some species of methodological misapprehension must be afoot. Some secret, sensible explanation of why this computational folderol works must exist, although its revisionary contours may look very different to what our engineers imagined when they first "justified" their reasoning policies through analogy to structural analysis. Indeed, in the 1950s applied mathematicians directed their attention to this foundational dilemma and located the requisite missing machinery within the functional analysis toolkit cited in the quotation from Lebedev et al. As J. T. Oden observes, the pioneering aircraft designers had stumbled across an excellent computational opportunity within the wrong "garden":

> *One unfamiliar with aspects of the history of finite elements may be led to the erroneous conclusion that the method ... emerged from the growing wealth of information on partial differential equations, weak solutions of boundary-value problems, Sobolev spaces, and the associated approximation theory for elliptic variational problems. This is a natural mistake, because the seeds for the modern theory of partial differential equations were sown about the same time as those for the development of modern finite-element methods, but in an entirely different garden.*[18]

We'll later learn that my Physics 101 instructor was cultivating essentially the same patch of deceptive landscape as our aircraft engineers.

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Developmental episodes like this—I call them *semantic mimics*—occur frequently in the history of human thought, when fruitful methods are discovered cloaked within misleading justificatory trappings. Learning
p. 336 how our aircraft engineers' finite element ↳ misadventures can be recast in more up-to-date terms should vividly illustrate the crooked pathways that developing patterns of improved reasoning sometimes follow.

To simplify our discussion, I will replace our engineer's little elastic plates with similar squares of membrane, even though the former are more prone to infinite stress problems than the latter. I do this largely because we can more readily appreciate the difference a dimension makes if we study essentially the same interior equation in both cases (viz. $kd^2z/dx^2 = f$ for strings and $k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f$) for membranes). Structurally, we might observe that circular membranes (such as a banjo head) don't present us with any corner singularities. Unfortunately, we can't decompose—the official term is tesselate—an arbitrary shape into circles without a lot of gaps. Through this decompositional doorway, squares and triangles insinuate themselves into central mechanical prominence, despite our initial impulse to dismiss their oddities as the result of excessive idealization.[19] We can't remain so haughty. We require their tesselation labor and must somehow reach a better reconciliation with their descriptive demands.

(iv)

The chief lesson instilled in the pages ahead echoes similar themes in our other essays, that the descriptive devices we initially select within applied mathematics frequently fail to match their targeted physical processes adequately and must be revised through sophisticated mathematical techniques. In particular, we must utilize the machinery of functional analysis to mollify the stand-alone, Sheriff of Bullet Valley pretensions of our interior membrane formula $(k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f)$ by borrowing characteristics from its partners in physical description (such as the manners in which we register data with respect to the boundaries and interfaces pertinent to a system of interest).[20] I use the phrase *descriptive harmony* to encompass holistic repairs of this type.

The seemingly anodyne philosophy of science primers that my brother brought home did not prepare me for these corrective necessities at all. Quite the contrary, they promote a coarse parsing of scientific discourse that
p. 337 encourages an exceptionally pernicious ↳ form of single-sentence idolatry, while sweeping into undifferentiated piles many of the descriptive ingredients we need to recognize and distinguish in order to effect the required repairs. In particular, these books write about "laws of nature" and "auxiliary conditions" in a manner that can't withstand a minimal degree of formal scrutiny (the "laws of nature," of course, are the parts we are supposed to idolize).

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single formula worship

Although received opinion within the philosophy of science has considerably altered since the heyday of the logical empiricists who wrote my brother's textbooks (Carl Hempel, Israel Scheffler, Rudolf Carnap, Hans Reichenbach, etc.), I find that little corrective effort has been applied to the descriptive inadequacies of which I complain. Modern writers continue to prattle about "laws of nature" and "auxiliary conditions" in the amorphous manner of those old books. If anything, the subsequent talk of "nomologically possible worlds" favored by David Lewis and his followers has only rendered the situation more opaque. Throughout this collection, I use the phrase "Theory T thinking" as a shorthand label to cover these common failures of diagnosis; the distinguishing ingredients within commonplace varieties of physical explanation have not been anatomized in adequate detail.

For example, here's how Ernest Nagel explicates the distinction between "laws" and other descriptive material, which he inexplicably calls "initial conditions":

> *Accordingly, a deductive scientific explanation ... must satisfy two logical conditions. The premises must contain at least one universal law, whose inclusion in the premises is essential for the deduction of the explicandum. And the premises must also contain a suitable number of initial conditions.*[21]

How an otherwise careful author could write so loosely confounds me, although allied mischaracterizations mar every specimen of logical empiricist explication I've surveyed. Why has Nagel employed the mathematician's term "initial condition" to embrace a wider motley of descriptive ingredients, when the term properly signifies "data pertinent to events occurring at a single moment t_0"[22] As explained in Essay 1 (I repeat the illustrative diagram here), neither boundary nor interfacial conditions operate in this manner at all; they
p. 338 both represent data laid upon surfaces that extend into the future. Nor are they always "singular" in any ↳ evident sense. In the bubble case illustrated, a standard boundary condition supplies a rule for how the inner gas diffuses through the bubble wall, a rule that will typically be articulated in the form of a differential equation involving a quantification over "all times." As Essay 1 also explains, in real-life practice boundary and interfacial conditions reflect special physical opportunities where pertinent data can be specified in cruder (and generally lower-dimensional[23]) terms those that we apply to the bulk material within the target system's interior (or its smooth sections, in the case of an interface). Confusing these specific forms of physical description with initial conditions in the proper sense automatically seals off any hope of understanding the forms of descriptive harmonization we shall investigate.[24]

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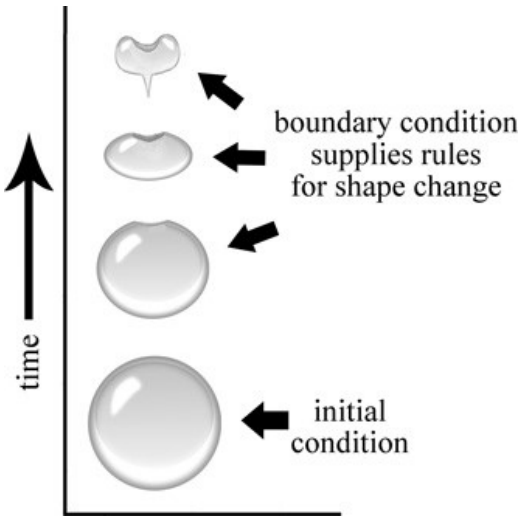

boundary conditions for an evolving bubble

One of the chief objectives of this collection of essays is to warn against the distortions of loose groupings such as this. A chief exemplar of "physics avoidance" that we visit often in different guises is the tactic of focusing upon the equilibriums of a target system rather than its evolving transitions (looking for Jack and Jill at the bottom of their hill rather than plotting their trajectories in getting there). In doing so, we avoid temporal consideration altogether, and so no notion of "initial conditions" enters our descriptive arena at all. But we will still employ laws (e.g. Poisson's equation) to map out the ambient gravitational potential. Why hasn't Nagel—or virtually any other philosopher writing in this vein—prepared us better for this adjustment in explanatory strategy, given its frequent appearance in scientific practice? And the same can be said for a large host of structural distinctions that are familiar to applied mathematicians but are completely ignored by the Theory T school.

There are some who praise coarse diagnosis as a philosophical virtue. David Armstrong writes:

> [Even if philosophers know no actual laws, they] *know the* forms *which statements of laws take ... It turns out, as a matter of fact, that the sort of fundamental investigations which we are undertaking can largely proceed with mere schemata of* [the sort "It is a law that Fs are Gs"] *... . Our abstract formulae may actually exhibit the heart of many philosophical problems about laws of nature, disentangled from confusing empirical detail. To every subject, its appropriate level of abstraction.*[25]

I seriously doubt that Armstrong could sort out the "laws" within a standard textbook according to any credible
p. 339 standard, and the "solution" he provides to "the problem of ↳ laws" is so "disentangled from ... detail" that one is completely unable to determine how it might reflect on any form of real life practice (e.g. our membrane formula or Poisson's law). The basis on which he believes that he can competently fix an "appropriate level of abstraction" without knowing anything about the subject so abstracted is beyond me. Throughout these essays we address many venerable philosophical puzzles about scientific method in (what I hope) will seem illuminating ways, relying upon the able guidance of standard applied mathematics distinctions. But Armstrong has decided on a completely armchair basis that none of that finicky stuff is required for his "philosophical purposes."

Why have such poor levels of philosophical analysis become so entrenched? There are a number of contributing factors. The first stems from the lingering heritage of what we might call Newton's celestial mechanics paradigm. The mathematics required for modeling the solar system in his manner proves especially simple from a conceptual point of view. We build up simple 3N sets of ordinary differential equations (ODEs) for a collection of N point masses entirely constructed upon the second law of motion frame "$d^2\theta/dt^2 = \Sigma f_j$,"

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where each θ represents the mass m_i of a component particle multiplied by one of its three Cartesian position coordinates (y_i, say). Each constitutive force summation Σf_j involved proves equally simple; one merely adds up the vectorial terms supplied by the law of gravitation $Gm_i m_j/|r_i - r_j|^2$ for every particle j other than particle i. The formal upshot is a collection of evolutionary ODEs that (except for some rare blow-up singularities) supplies us with a completely determinant solution over time once an adequate set of initial conditions—positions and velocities for each particle and each Cartesian direction—have been provided for a starting time t_0. We typically approach our solar system as an "island universe" in the sense that no stuff lying outside our closed set of point mass representations is considered. As a result, no boundary conditions whatsoever enter the descriptive arena.

solar system

Even with respect to this most favorable setting, deep ambiguities in standard Theory T analysis emerge. Should the celestial mechanics ODEs thus constructed qualify as "laws" themselves or merely some of the ingredients from which they have been composed (i.e. the Newtonian **F** = m**a** scheme and the law of gravitation)? The number of objects in the universe, their respective masses, and the fact that they only interact via gravitation don't strike most of us as "laws" in any apparent sense, but we can't construct our ODE sets without relying upon these supplementary facts (which might be plausibly labeled as "modeling conditions"). But we plainly require "universal generalizations" of ODE strength to move our target system forward in time from its initial conditions in the manner that Nagel suggests. So what, within this most favorable exemplar of modeling circumstances, qualifies as a "law"? Our philosophy of science primers rarely tell us and wobble disconcertedly between settled readings.

p. 340 As soon as we leave the preserves of celestial mechanics, "modeling circumstances" within physics rarely follow such a simple descriptive recipe.[26] Complications arise as soon as we consider interactions between matter other than gravitation, such as the bonds that hold our planets and sun together. Our **F** = m**a** recipe anticipates that physics will produce additional Σf_j terms to capture these factors, but, in point of fact, classical mechanics practice never developed a canonical view on what these missing "laws" should look like. Instead, applicational practice changed the subject by shifting to modeling frameworks of a markedly different character, such as puffing up the erstwhile point masses into either rigid bodies or the flexible blobs of continuum mechanics. Both of these frameworks require more elaborate behaviors from matter than our **F** = m**a** basis did, and a proper treatment of the blobs requires a shift to partial differential equations (PDEs).[27] Under these shifts in mathematical setting, all of the suppressed-within-celestial-mechanics aspects of physical modeling flood back onto the stage: finite dimensions, boundary and interfacial conditions, constraints, Fourier decompositions, etc. My chief plea throughout this book of essays is that we warmly acknowledge the substantive and individualistic contributions that all of these ingredients contribute to the elaborate pageant of scientific description.

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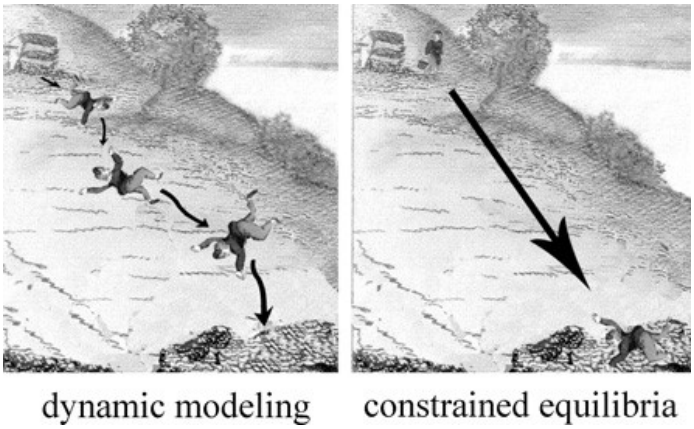
Historians of mathematics often comment on the gradual manner in which mathematicians slowly realized that PDEs and their more complex descendants represent subtler critters than their evolutionary ODE forebears (they labored for a long time under the "illusions of ODE expectation"[28]). Although mathematicians have since learned better, many philosophers have not. Insofar as I can determine, none of the old ODE-generated misapprehensions have been corrected within standard Theory T thinking. In fact, the latter has made matters worse by recasting the reasoning patterns pertinent to an ODE equational basis into the format of elementary logic. It accomplishes this trick by converting the smoothly flowing stages of an ODE-driven evolutionary process into long chains of modus ponens instantiations based upon alleged "general laws" such as "At all
p. 341 times t, if conditions A hold at t, then condition B will hold ↳ at time $t + \Delta t$." As Essay 2 explains, use of the latter idiom should be regarded as merely a time-stepped surrogate for a type of causal process that is conceptually better encapsulated within an evolutionary set of differential equations. But if we accept this replacement without cavil, we are likely to presume that the main instruments of inferential advance within mathematical physics are supplied by the rules of elementary logic, rather than the skills involved in seeking out helpful inferential adjuncts such as Green's functions. In the history of science it is not uncommon to find important notions entering the descriptive arena in the initial guise of approximation statements (which is what a finite difference scheme represents), which are later corrected to more accurate formulations as expressive facility within mathematics improves. Sometimes a considerable span of time is required before the true novelty of the new procedures is fully recognized and placed upon an accurately diagnosed mathematical basis. This, in a nutshell, is what eventually happens to the little "plates" of our aircraft engineers, as we'll learn in the details to follow.

But the diagnostic practices of the Theory T school operate in a contrary manner—they wash out crucial differences in explanatory tactics by reformulating everything within their favored "for all times t, if conditions A hold at t, then condition B will hold at time $t + \Delta t$" format. As Essay 2 argues at length, we do not want to confuse predictive policies that track an unfolding dynamical process through evolutionary equations (such as Jack tumbling down the hill on the left of the illustration) with the equilibrium-based recipe on the right (which searches for Jack's bottom-of-the-hill equilibrium position without engaging in any evolutionary plotting whatsoever). But Theory T thinking only requires a simple "spatial step Δs for time step Δt" exchange to blur the two explanatory formats together. Dressed in similar logistical cloth, such tailoring makes an equilibrium state calculation (where will Jack land?) *look like* one designed for an evolutionary process (what is Jack's trajectory down the hill?). Pace Armstrong, melding these distinct methodologies together advances no legitimate intellectual purpose, for one of philosophy's chief tasks is that of untangling conceptual confusions, not adding to them.



distinct modeling strategies

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Hadamard often stresses the fact that we can't really understand the explanatory purposes of a set of differential equations unless we appreciate how they should fit together with their anointed side conditions (viz. initial and boundary region values, interfacial jumps, and so forth). Cramming all of these important considerations into an undifferentiated grab bag labeled "auxiliary conditions" (or, heaven forbid, "initial
p. 342 conditions") deprives us of the very discriminations we require for unraveling many ↳ characteristic forms of conceptual puzzle.[29] Indeed, the basic mission of this essay is to underscore this lesson, by showing the oddities of the little "plates" of the airplane designers traces to an unexpected manner in which various side condition ingredients secretly cooperate with the central modeling equations. Closely related considerations will help us unpack the "impossible things before breakfast" mysteries of Physics 101 instruction. In both cases, we confront scientific illustrations of our Sheriff of Bullet Valley theme: The helpful nephews that secretly prop up "I do it all alone" pretensions of Theory T's "laws of nature" turn out to be boundary conditions, interfaces, constraints, and allied auxiliary ingredients.

(v)

With respect to the solitary formula worship of "laws of nature," let us again review J. N. Reddy's warning, already cited:

> *It should be understood that the equations governing a physical problem are themselves approximate. The approximations are introduced through several sources, including representation of the geometry and boundary conditions, loads and material behavior.*

Observe that his "several sources" include the very modeling factors generally lumped together by Theory T thinkers. Note as well that Reddy regards the interior equations (such as our membrane formula $k(\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2) = f$) as "approximate" in the same manner. He offers the sound advice that, when descriptive disharmonies or other oddities arise within our descriptive policies, we should look to the mathematical characteristics of all contributing parties—including our interior equation "laws"—for evidences of descriptive overstatement. This is because suitable correctives often involve some shuttling of informational content between our erstwhile characterizations of "boundary," "interior," "interface," and so forth. Behind all of this is the holistic viewpoint articulated by Lebedev and his comrades: well-wrought descriptive ingredients must work together as a collective team. Any allocation of a local task (such as "describing an interior region") must accord harmoniously with the divisions of labor undertaken by other descriptive ingredients. If "boundaries" and "interiors" don't match up in suitable ways when formulated naïvely, their respective contents need to be readjusted until a better accord is reached.

These methodological morals are encouraged by the scale-centered considerations of Essay 1. There we noted that the information registered within an "interior equation" and a "boundary condition" typically follow different drummers; we average (or "smear") over fine details in the first instance and *anti*-average (or
p. 343 condense) data in the second. We further observed that these informational allocations are apt to alter as ↳ we move from one RVE cutoff scale to another and that the mathematical tools employed to these purposes sometimes need to be rather sophisticated.

But how do we achieve these improved descriptive harmonies mathematically? Rather than directly pursuing the problems of harmonizing square boundaries with uncooperative interior equations (which would led us to Green's functions of "boundary element" type (type (2) above)), let us shift to allied interior harmonizations of an "influence function" type (type (1)), where matters are expositionally easier.[30] Indeed, concerns of a Green's function ilk arose within mathematics long before Green himself arrived on the scene, for they emerged immediately in conjunction with the first proper example of a partial differential equation in the history of physics, the one-dimensional wave equation (frequently discussed elsewhere in this book in application to

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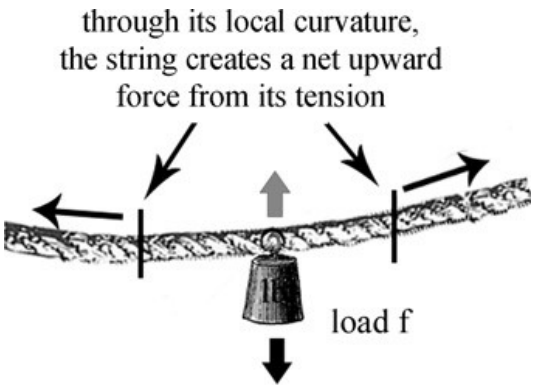
violin strings). This formula formed the core of a celebrated conceptual dispute between d'Alembert and L. Euler with respect to its intended "contents."[31] In that original setting, Euler and d'Alembert were concerned with a Green's function in the propagator sense ((3) above), but, for simplicity's sake,[32] I will adjust their discussion to the static circumstances of a stationary string loaded with weights, exactly in the form that I encountered in Physics 101 long ago. Here the pertinent interior formula is $k\ d^2y/dx^2 = f(x)$, where f captures the local gravitational force density weighing the string down at position x. In plain English, this formula expresses the equilibrium requirement that, at each interior point x within the string, the local string curvature (measured by d^2y/dx^2) should remain proportional to the downward gravitational pull of the load f(x) at that same position. In other words, taut strings support weights by curving an appropriate amount determined by their elasticity. If our string is unevenly loaded—i.e. the load function f(x) is irregular—then its sagging shape y(x) may outline a very complex curve. We might observe that d'Alembert's original derivation of this equation relied upon many "impossible things" as discussed in section (ii). Our eventual review of their dispute will consider a more rational means of reaching this same equation.



setup element for a loaded string

p. 344 Euler offered the following observations with respect to this modeling. If we load our string at a single position α with a unit weight completely localized to α, the resulting sagging should intuitively produce the "V_α"-function curve illustrated. Accordingly, $V_\alpha(x)$ seems to capture the influence that a single unit weight hung at point α will have on the rest of the string. Because our equation is linear (= its solutions superimpose), we can calculate the combined effect of a smeared-out array of weights f(x) by simply adding up a weighted average of the various $V_\alpha(x)$'s corresponding to each local α (more exactly, $y(x) = \int f(\alpha)V_\alpha(x)d\alpha$). Euler thereby emphasizes the basic considerations that allow his $V_\alpha(x)$'s to serve as one-dimensional Green's functions for the interior of his string, completely comparable to the two-dimensional influence functions we assigned to a taut membrane under fixed boundary moorings.

fundamental solution for loaded string

But d'Alembert immediately objected to Euler's observations in the prickly manner familiar to original authors (paraphrasing): *Wait a minute! My derivation of that formula requires that the strings they govern must possess clearly defined curvatures—that is, d^2y/dx^2—everywhere along their lengths. I obtained that formula by working with legitimate forces in an '$F = ma$' manner. But the sharp corner in your $V_\alpha(x)$ changes too abruptly to possess*

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even a first derivative (dy/dx)*. So how can your sharp-cornered proposal be regarded as implementing the physics I captured within my original modeling equation?* Here d'Alembert poses an entirely sensible question and the answer we shall develop will prove unexpectedly subtle.

d'Alembert and Euler

Here's a more modern way of articulating d'Alembert's concerns. In response to the applied weights, a string stores strain energy through curving—that is, it stores a capacity for performing work that allows it to return to its natural uncurved state. The more our string is curved, the more its store of internal energy increases. If we somehow manage to bend our string into Euler's tight corner, we must have stored an infinite amount of energy in that singular location.[33] But a singular point that contains an infinite store of energy is probably mechanically incoherent and was surely not contemplated in d'Alembert's original assessment of the ↳ physics relevant to a string. Plainly, Euler has proposed a one-dimensional analog to the impossibly concentrated energies in the corner's perfectly square elastic plates.

p. 345

And so matters stood for many years. Applied mathematicians continued to avail themselves freely of Euler's $V_\alpha(x)$ as an important inferential aid, while simultaneously characterizing these invocations as "unrigorous," fit for mathematical consumption only when housed inside an integral expression as a "kernel." Modern mathematical thought views this diminishment of $V_\alpha(x)$'s inferential labors as unjust, the result of assigning the interior formula $k\ d^2y/dx^2 = f(x)$ an excessive degree of stand-alone descriptive credit. Properly regarded, applied mathematicians see the string equation as operating in collective harmony with the relevant boundary conditions (here: the assumption that the string stays rigidly moored at pin and bridge). If our present parsing of "$k\ d^2y/dx^2 = f(x)$"'s informational content does not provide a harmonious match up with Euler's $V_\alpha(x)$ characterization, then we must rethink what "$k\ d^2y/dx^2 = f(x)$" signifies from a physical point of view. So mathematicians twiddle with its traditional "assigned meanings" until they reach better accord with the string's anticipated loads and boundary conditions. How Laurent Schwartz and S. I. Sobolev[34] performed this semantic rebalancing in the mid-twentieth century will comprise the expository topic of section (vi).

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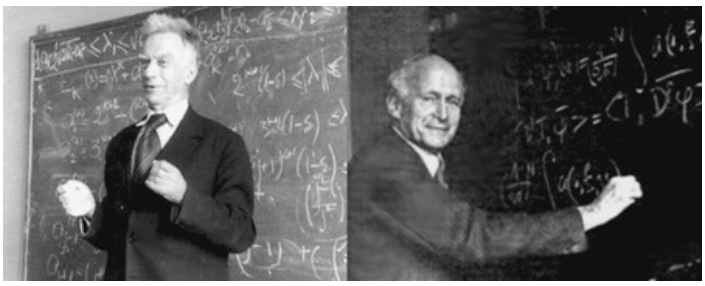

Sobolev and Schwartz

Without fully recognizing the fact, the earlier mathematicians who imprisoned the legitimate employments of $V_\alpha(x)$ within the "kernels" of integral expressions displayed a dim awareness of the cooperative symbiosis that should obtain between interior and boundary descriptions, because a Green's function such as $V_\alpha(x)$ acts as a natural bridge between our problem's interior conditions and its boundaries. In that sense, they had already discovered most of the rules that determine how $V_\alpha(x)$ should behave; they were merely packaged in disguised form. Our main task is to assign both "$V_\alpha(x)$" and "$k\ d^2y/dx^2 = f\ (x)$" revised semantic meanings that remove the inequities of this unnecessary imprisonment within a kernel. These are the circumstances that Jean Dieudonné has in mind when he comments:

> *[M]ost of the problems which belong to the theory of distributions had been considered and essentially solved before Schwartz, but no one had succeeded in building up a formalism which would dispense of special arguments in each particular case.*[35]

p. 346 Essay 1 stresses the important role that contextual embedding plays in crediting linguistic expressions with apt measures of informational content. Locking our $V_\alpha(x)$'s inside integrals is a crude way to implement a required dependence upon wider context. So let's see if we can't perform the same chore in a more elegant manner.

(vi)

A persistent methodological theme that runs through the long history of applied mathematics is the recognition that the readily available tools of pure mathematics (e.g. functions straightforwardly composed from elementary operations) rarely capture targeted physical behaviors accurately over wide expanses of space and time, but fall out of accurate correlation and must be correctively repaired somehow (e.g. by allowing a non-analytic join between two sectors described by analytic functions). By the mid-nineteenth century, these regenerative needs became so pronounced that novel policies for fashioning fresh domains of unexpected mathematical objects had to be found, frequently involving large dollops of set theory-style thinking. In the 1890's, Hilbert proposed a novel method for resolving some harmonization problems that had popped up within many disparate branches of mathematics, sometimes in severely damaging ways.[36] It will help us appreciate Schwartz' later proposals for dealing with $V_\alpha(x)$ if we first look at Hilbert's characteristic approach to an allied problem. One of his chief arenas of concern arose in conjunction with the problems of optimal planning. Suppose that there's a river island that we'd like to reach from a fixed point on the shore and we're concerned to minimize our rowing effort in terms of the length of the journey and the energetic effort required to turn our boat in mid-stream (which is a function of the curvature of the path). A dock intrudes into the river that we must circumnavigate as well. If the water is slower near the shore, it can easily happen that the optimal rowing plan intuitively assumes the form of a so-called "bang-bang solution": head straight towards a spot on the far shore and then turn back rectilinearly to the island. If we explore the "space" of curving paths that reach

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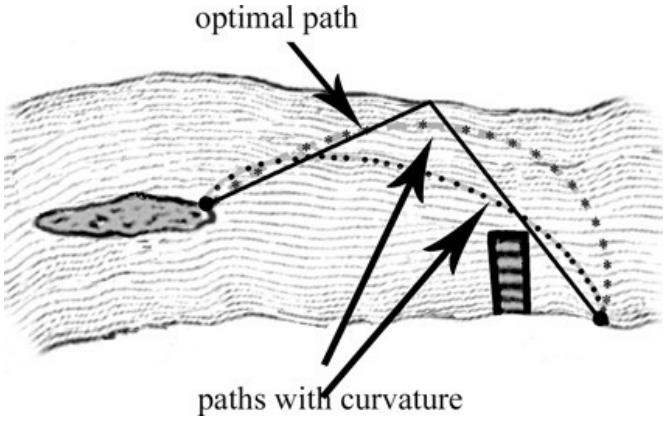
p. 347 the island by other routes, we find that all of these alternatives require a ↳ greater degree of rowing effort and surround our optimal bang-bang sailing path as a fixed point. So what's the problem?



an optimal path problem

Well, we're confronted with a situation that essentially reproduces all of d'Alembert's worries. We'd like to claim that our favored path $\mathscr{P}_{\text{optimal}}$ involves a minimum amount of rowing energy, but a vital component within any energy budget depends upon the curvature $\left(= d^2y/dx^2\right)$ everywhere along the path. But our sharp turnaround point is not curved at all—so how can we assign a coherent degree of "turning energy" to $\mathscr{P}_{\text{optimal}}$? In mathematical terms, we are looking at a situation where our bang-bang rowing scheme should comprise the greatest lower bound, measured energywise, to a set α (or "space") of curved paths that possess proper curvature everywhere. But our desired rowing function does not belong to the set α. In modern jargon, our space α is not complete with respect to its energy norm (= the rule whereby we assign total energies to possible rowing paths).

Hilbert's suggested cure—usually called Hilbert's direct method within the calculus of variations—was inspired by Richard Dedekind's policies for filling in the irrational "holes" in the real number system through suitable cuts (or allied constructions). Why not let the curved paths surrounding our optimal trajectory $\mathscr{P}_{\text{optimal}}$ install an extended energy measure upon $\mathscr{P}_{\text{optimal}}$ as a friendly, neighborly gesture? More exactly, if the ascertainable energies on all nearby curved paths near $\mathscr{P}_{\text{optimal}}$ converge to the value E_4 as their energetic bound, we should credit $\mathscr{P}_{\text{optimal}}$ with E_4 as its new "turning energy." Here's a homely analogy to illustrate the general idea. Suppose that our troops are commanded by a haughty general who is too stupid to know how to lead his troops wisely through the battlefield before him. Despite these flaws in intellect, he may still perform ably if he relies upon a cadre of aides-de-camp who understand the terrain better and can advise him, "Uh, better turn slightly to the left, commander." Just so: under Hilbert's plan, the space of smooth (but non-optimal) rowing plans enclose the desired rowing path within a topological mesh that transfers their $\partial^2 y/\partial x^2$-based understanding of what a normal turning energy should be into "advice" (= an energy norm that can be uniquely extended to a wider measure) appropriate to "bang-bang" controls as well. In other words, we complete the space of energetically measurable rowing plans by relying upon the wisdom of its non-optimal members, thereby extending the evaluative reach of our energetic cost norm to cover sharp-cornered rowing schemes as well.

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which way should I lead?

In adopting this plan, Hilbert institutes an important division of descriptive labor amongst the members of his solution space α. The completely curvy paths inside α ↳ know how to directly evaluate their own energy costs based upon how their derivatives behave, but often they can't supply us with a suitable optimal rowing path. In contrast, the broken line paths perform the latter task admirably, but they cannot evaluate their own energy costs directly. So they need a little help from their friends, the curvy paths, which they obtain by becoming trapped within a coercive topological mesh comprised of these comrades.

p. 348

For this policy to work out, we must build up the space α and its norm of closeness in a careful manner. If we allow too much extraneous junk in α or prove too cavalier in our standards of closeness, our bang-bang curve might become trapped inside a number of distinct function meshes that will assign it inconsistent energetic costs. This basic consideration—that useful spaces of possibility should be kept small and appropriately focused—comprises a central theme of Essay 7.

J. T. Oden and J. N. Reddy analogize these extension techniques to filling in the missing "pinpricks" within a space of functions:

> *Geometrically, the process of completion from $\boldsymbol{H}_2$ to $\boldsymbol{H}_1$ has a simple interpretation. The quadratic I is represented by a complex surface, a paraboloid in infinite dimensions. At first, when I was defined only for v in H_2, there were "holes" in this surface; all we have done is to fill them in. The surface has changed neither its shape nor its minimum value; it's just that the pinpricks, corresponding to functions v lying in H_1, but not in H_2, no longer exist.*[37]

Indeed, their $\boldsymbol{H}_1$ is the space in which we'll want $V_\alpha(x)$ to live and I will eventually supply it with an extended energy measure. But I'm getting slightly ahead of my story.

(vii)

To follow out the usual reasoning employed to motivate Schwartz's formal policies, let's return to the "descriptive injustice" latent in Barton's off-handed remarks on the relationships of "idealized" to "true" shapes of a plate:

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> *only in regions negligibly small by normal standards is there any appreciable difference between the solution calculated for the true shape with rounded corners and the idealized solution calculated for idealized rectangular shapes.*

But we have already observed that our usual modes of describing our plate's interior (using our membrane equation) enjoy no evident descriptive superiority over our usual techniques for characterizing its bounding edges, beyond the fact that the former is much larger in terms of area than the latter. Upon what empirical basis do we frame ↳ these descriptive judgments? Answer: from measurements within the two basic regions. In this vein, F. G. Friedlander ruminates:

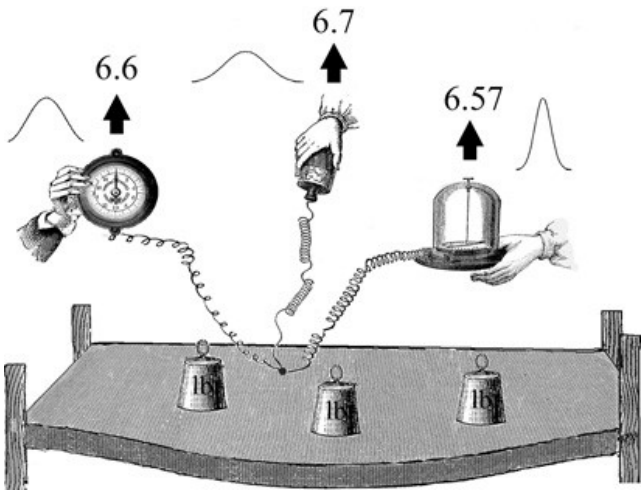
p. 349

> *It is usual to think of the physical variables as functions of position in space-time. But this is already a mathematical model: one never measures the pressure, for example, at a particular point and a given instant, but only an average over a small region and a short time-interval. The result of such a measurement can equally well be expressed as p. φ where φ is a test-function which represents the averaging of the pressure effected by the actual measurement. Thus a mathematical model in which the physical variables are represented by distributions and the observations by the linear functionals which these define is not necessarily more artificial than one in which the physical variables are represented by continuous and differentiable functions defined on a mathematical continuum.*[38]

In an allied framework, Robert S. Strichartz writes:

> *[I]t is often more meaningful physically to discuss quantities like If(x) θ(x) dx than the value of f at a particular point x. The secret of successful mathematics is to eliminate all unnecessary and irrelevant information ... Since we have decided that the value of f at x is essentially impossible to measure, let's stop requiring our function to have value at x.*[39]

What do these authors have in mind? They regard their test functions $\theta(x)$ as mathematical models of the measuring instruments we employ to probe a region of target system for its pressure or strain energy content. Because of their different internal workings (largely unknown to us), different measuring instruments θ and θ^* are likely to supply somewhat different readings when applied to the same regions, as illustrated. Why? Because measuring instruments inevitably smear out (the mathematician's official term is mollify) any local behavior into a broader response curve according to their internal construction. Accordingly, Friedlander and Strichartz recommend that we regard the properties of the plate itself in an operational manner, following the old adage "by their fruits you shall know them."



test function probes

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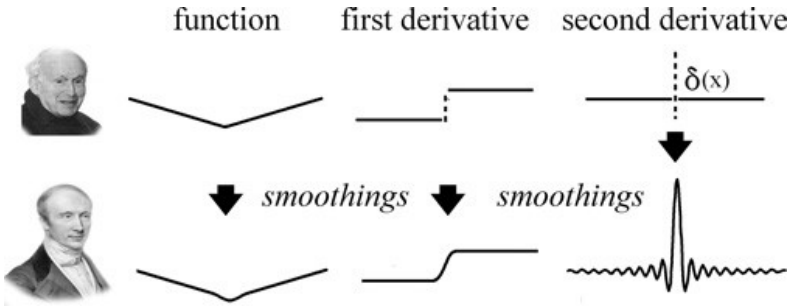
In mathematical terms, this means that our erstwhile "strain energy" property is now modeled as a functional F that operates on a measuring instrument function $\theta(x)$ to supply a numerical assessment of strain energy
p. 350 content $F(\theta(x))$. From this point of view, ↳ two regions of surface contain the same strain energy content if they affect all measuring instruments in the same way, an assertion that we can now apply to modeling gizmos[40] like our sharp-cornered Green's functions despite the fact that they lack the derivatives required to possess a "strain energy" in the old-fashioned, unenlightened manner.

In fact, Schwartz reworks these basic ideas in a clever manner so that our various Green's functions turn out to represent genuine solutions to their parent problems once "solution" is reinterpreted in a suitable manner. The trick is to also rework the notion of "derivative" so that our sharp-cornered functions pick up derivatives from their smoother functional companions. We shall illustrate how this works with $V_\alpha(x)$. Once again, the basic strategy is to peek at the copybook of various approximations lying near to $V_\alpha(x)$.

peeking at nearby answers

To this end, Schwartz sets up a special space of test functions **D'** comprised of completely smooth functions θ that he will employ to smear out $V_\alpha(x)$ enough that the mollified results will possess regular calculus derivatives. He does this because he wants to view our target function $V_\alpha(x)$ through the lens of how it appears to the measuring instruments modeled by the θ's within **D'**. So he treats $V_\alpha(x)$, not as a regular function, but as a functional that acts on all of the test functions θ and produces a mesh of smooth curves that surrounds the two sides of the erstwhile function $V_\alpha(x)$ as closely as we might like. Intuitively, the "derivative" that we would like our functionalized $V_\alpha(x)$ to possess should be the broken-line function ____- - - -, indicating that $V_\alpha(x)$ possesses a constant negative slope up to the point α and a constant upward slope thereafter. How does Schwartz persuade the functionalized version of ____- - - - to serve as the first derivative of $V_\alpha(x)$? Answer: by asking all of the regular calculus derivatives of the θ-smoothed companions of $V_\alpha(x)$ to surround the two pieces
p. 351 of ____- - - - in a similar mesh. If we repeat this demand a second time, we obtain a functional second ↳ derivative for $V_\alpha(x)$ that isn't similar to a normal function at all, but rather represents a Dirac δ "function" blip.



smoothing a function so the calculus can understand it

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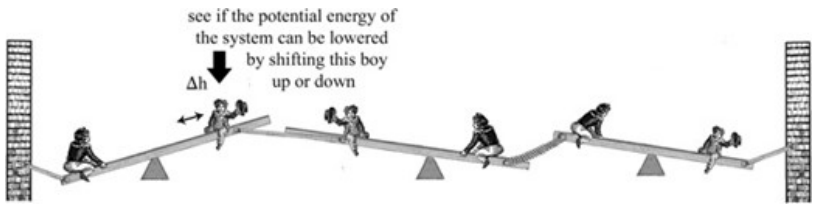
The upshot of all of this is that our formerly out-of-territory expression $V_\alpha(x)$ is now welcomed as a straightforward new solution,[41] under our new reading of "d^2y/dx^2 '" to $k\ d^2y/dx^2 = f(x)$, in the circumstances where $\delta(x)$ serves in the f(x) role as a loading "function." In this way, our erstwhile extension-element "fundamental solutions" now qualify as genuine "solutions" of an extended tolerance. Welcome, previously excluded comrade, to the fold!

So this is how we should parse the claim that the string is loaded by a unit weight completely localized to the point p—the string as a whole is characterized by a Dirac functional δ centered at p which supplies a non-zero output $\delta(\theta)$ only on measurements θ that cover the point p.

By proceeding in this general manner, we can fill in many of the other pinpricks of non-harmonization between possible loads and their interior solutions. The general key to these improvements is the sharing of tasks. We ask our helpful cadre of adjutants (i.e. functionals mollified by θs) to evaluate circumstances on our behalf (what should my d^2y/dx^2 look like?) that we can't directly evaluate for ourselves. Paraphrasing Essay 1 again, in union there is strength.[42]

(viii)

As a consequence of these procedures, even perfectly regular solutions g(x) to $k\ d^2y/dx^2 = f(x)$ (i.e. solutions satisfying old calculus rules with a smoothly distributed f) become likewise surrounded by a dense swarm of approximations, including a lot of adjutants with sharp corners in them (whose relationships to g(x) are monitored by their θ-function smoothings). This surrounding swarm allows us to make an unexpected connection with a venerable theme in mechanics much discussed in Essay 7, the notion of virtual work. Without fully regurgitating the discussion of that essay, we merely note that an excellent way of characterizing the equilibrium states of a loaded static system is to consider whether its stored energy can be lessened by
p. 352
varying the parts of some initial trial guess G_0 in various "virtual" manners. If a wiggling to G_1 ↳ improves matters, consider G_1 to be our new guess and test whether G_1's energy storage can be improved to a better G_2. If circumstances are favorable, the resulting string of ever-improving guesses $G_0,\ G_1,\ G_2 \ldots$ will converge to a final fixed point G_4 that represents our equilibrium state. In the illustration, at equilibrium our complex assembly of children-laden seesaws will settle into one of the zig-zag configurations Z(x) that represent the lowest energy state available to the system, a feature that we characterize in variational terms by observing that any attempt δh to raise or lower one of the children from his position at h will prompt a countering response that pulls the device back towards h.



searching for an equilibrium balance

These virtual work techniques can be applied to a continuous system such as our loaded string, in which circumstances the variations upon our children's elevations turn into alterations of a trial solution y(x) employing a varied function $\delta y(x)$.[43] Employing these ideas, we can reformulate the physical demands placed upon a solution by our old friend $k\ d^2y/dx^2 = f(x)$ with the more generous reading:

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> $\int(\partial y\,(x)/\partial x).\,(\partial\delta y/\partial x) = \int f\,(x, y)$ for all variations δy's within a suitable manipulation space **M** and where y(x) belongs to a space of permissible configurations **Z**.[44]

Note that these integrations stretch across the full length of our string, including its endpoint boundaries. This revised formula is usually called a weak reading of the originating equation $k\ d^2y/dx^2 = f\,(x)$. For various reasons to which we'll later return, such recastings are often regarded as a superior means of expressing the basic principles of mechanics, a point of view endorsed by Lagrange within his celebrated Analytical Mechanics and echoed by Oden and Reddy as follows:

> *[T]he importance of variational statements of physical laws, in the general sense of these terms, goes far beyond their use as simply an alternative to other formulations. In fact, variational or weak forms of the laws*
> p. 353 *of continuum mechanics may be the only natural and ↳ rigorously correct way to think of them. The idea that they are only equivalent substitutes for local statements of these laws is an all too common misunderstanding. The fundamental principles of mechanics are global principles; they may require local integrability of certain fields, but not local differentiability. Hence we can generate local forms of these laws only if we endow physical field quantities with a possibly unnatural degree of smoothness. This done, we rule out all traces of point sources, discontinuities, and their derivatives, and we restrict ourselves to a rather unrealistic view of the universe.*[45]

Conceptually, a manipulation on a target modeling is quite different from the modeling itself, just as a measuring instrument is obviously distinguished from the target to which it is applied. Nonetheless, for inferential purposes it is often convenient if we can persuade mathematical representatives for these conceptually distinct notions to live within a common arena. The standard illustration of the merits of cohabitation can be found amongst vectors and their dual one-forms. Consider the contour lines on a hillside. If a traveler moves along a certain vector V in a brief amount of time, her potential energy will increase or decrease according the number of contour lines her path transverses. If we look at these circumstances locally, these contours can be adequately approximated by a set of equally spaced lines and our traveler's displacement by a short straight line vector V. Conceptually, we can model these circumstances in the following way. A gravitational one-form G (= a function on vectors) inspects the traveler's V and produces a numerical evaluation n (= the change in potential energy along that path, according to the number of local contour lines that V falls across). In other words, $G\,(V) = n$. According to this point of view, G lives in a different space (the cotangent space) than the V's, which dwells in the tangent space $\mathfrak{S}_V$ to the underlying manifold (their pairing is usually called a Banach space). But once we have a metric operative in S_V, we can push a natural representative for G into $\mathfrak{S}_V$ and operate entirely within $\mathfrak{S}_V$. Indeed, a suitable representative for G lies readily to hand: the gradient vector that marks the direction of largest potential energy change available from our traveler's
p. 354 starting point. This allows our original "across different ↳ spaces" computation G(V) to be replaced by a wholly inside $\mathfrak{S}_V$ surrogate: the projected length of V onto grad(G), otherwise marked as the inner product (V, grad(G)). As noted before, this push forward of G into $\mathfrak{S}_V$ proves computationally advantageous whenever the strategy proves viable, but does not alter the fact that gravitational contour lines and displacement vectors are conceptually quite different. Historically, mathematical understanding unraveled in the opposite direction: G and V were both originally modeled as vectors and the intellectual probity of E. Cartan and others pulled the two notions apart later on (up-to-date calculus classes now teach students in Cartan's manner). A similar confusion of roles plays an important part in the Physics 101 saga we are finally prepared to unravel.

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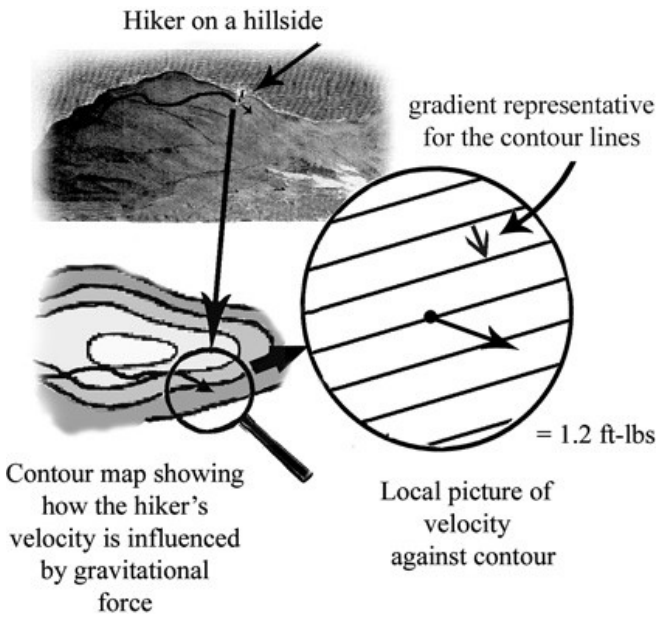
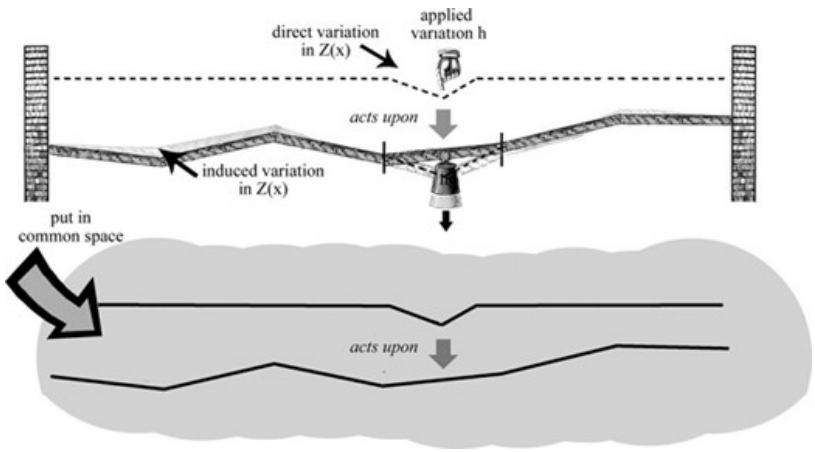


one-form acting on a vector

In the context of our virtual work considerations, combining trial functions and their potential manipulations into a common space isn't usually hard, for both ingredients are naturally modeled as functions obeying the same endpoint conditions (= remaining statically fixed in the cases we have considered). But I again stress that "identical natural modeling" needn't connote conceptual identity of roles—a manipulation isn't the same thing as a trial solution. Within this commodious combined setting, our previous virtual work manipulations can now be registered as simple inner products (as $(\mathrm{ky}(\mathrm{x}),\ \delta\mathrm{y}(\mathrm{x})))$. If we now throw Schwartz's extension elements into our mix, we find ourselves with enlarged collections of both trial solutions and possible manipulations that include disjointed specimens of a $V_\alpha(x)$ character, that is, variational functions that possess derivatives only in Schwartz's extended sense.



combining manipulations and trial functions

(ix)

Once we locate ourselves within this harmoniously enlarged setting, a remarkable computational opportunity comes into view. Although a true variational solution to our weak string equation under a normal distributed loading usually emerges in a fixed point limit as a completely smooth function y(x), we now have the

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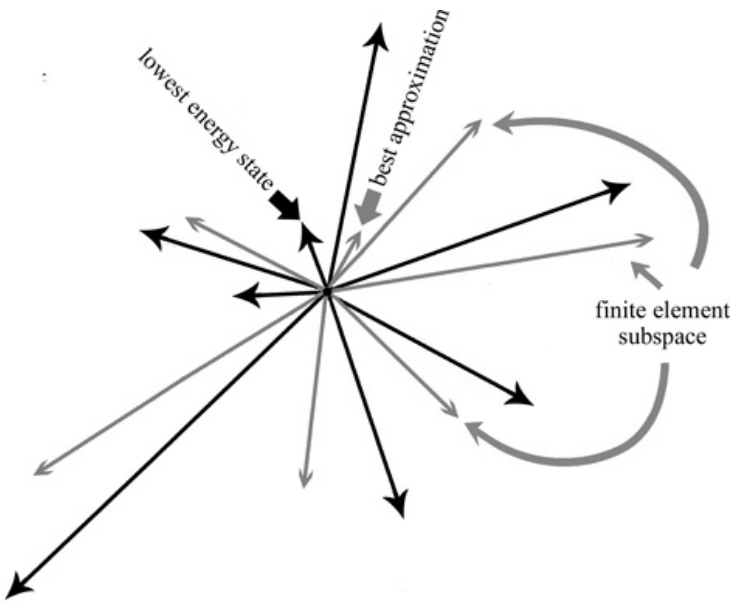
p. 355 conceptual resources ↳ to ask: "within a suitable subspace $\mathfrak{S}_{\mathrm{FE}}$ of broken-line approximations to y(x), which of its trial functions will wind up possessing the least amount of energy as we tweak their nodal heights up or down in basic children-on-a-seesaw fashion?" This answer will provide us with the best approximation $y(x)_{\mathrm{FE}}$ to our equation's true solution available within our smaller subspace $\mathfrak{S}_{\mathrm{FE}}$ of broken-line proposals. Computational techniques that follow this "search within a subspace" tactic are said to follow the Rayleigh–Ritz method, which, courtesy of Schwartz's innovations, can now be extended to include broken-line approximations. Because our problems are linear, we can locate these best $y(x)_{\mathrm{FE}}$ subspace answers by first looking at variations localized to small sectors of our string and then adding these up (just as we only wiggle one child at a time when we search for the equilibrium of our linked seesaws). If we limit our approximation subspace $\mathfrak{S}_{\mathrm{FE}}$ to simple broken-line functions only, we find ourselves looking for the sum of hat functions that most closely approximates our string according to the Rayleigh–Ritz principle.[46]



Rayleigh–Ritz principle

An additional advantage emerges when we also restrict our inner product computations to our approximation subspace $\mathfrak{S}_{\mathrm{FE}}$; the responses to variations reported in our original weak equation convert to simple algebraic requirements on the manner in which the nodal heights of adjacent elements in our grid must relate to one another. These internodal requirements are easy to implement on a computer, allowing the device to quickly find the best broken-line approximation to our lowest energy demand available within the specific $\mathfrak{S}_{\mathrm{FE}}$ space employed. It is through recasting our original physical problems into this enlarged and better harmonized setting that applied mathematicians now locate the basic strategic underpinnings of the various finite element methods widely utilized in computational practice today.

From a philosophical perspective, the most striking aspect of this saga lies in the fact that our original aircraft
p. 356 engineers stumbled upon the rudiments of this technique ↳ following a completely erroneous picture of what they were doing. William Strang and George Fix summarize the historical situation as follows:

> *Unhappily none of the credit for* [the finite element] *idea goes to numerical analysts. The method was created by structural engineers and it was not recognized at the start as an instance of the Rayleigh–Ritz principle. The subdivision into simpler pieces, and the equations of equilibrium and compatibility between the pieces, were initially constructed on the basis of physical reasoning. The later development of more accurate elements happened in a similar way; it was recognized that increasing the degree of the*

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> *polynomials* [in the sections of the trial functions] *would greatly improve the accuracy, but the unknowns q_j computed in the discrete approximation have always retained a physical significance.*[47]

By "physical reasoning," they mean reliance upon the supposition that we might manufacture an airplane wing by welding together little squares. As we saw, the starting point of the aircraft engineers began in the supposition that the algebraic rules employed in a finite element computation represent simplified Green's function accounts of how adjoined plates affect one other across their fused boundaries. Such a story is implausible; one generates very high stresses if one twists individual plates in that manner. The notion that the strong boundary region characteristics of welded component plates will somehow mysteriously vanish if one makes the plates small enough is physically ridiculous. Nonetheless, this supposition is the exact analog of my Physics 101 instructor's blithe contention that any contributions from the glued-together portions of his segmented string will likewise disappear in some vaguely designated "limit." No wonder that I, in the Theory T prissiness of my student days, was utterly bamboozled by such assurances.

But we now recognize that no dissection into physically meaningful segments plays any role within the finite element procedure, properly understood. Instead, the method represents a general approximation technique applicable to a wide variety of applications, opening up a wide computational opportunity that was immediately seized upon as soon as applied mathematicians recast the underlying rationale for such tactics in properly diagnosed terms. In retrospect, what went wrong in our aircraft engineer's original understanding of their methods? Answer: they were confused by the computational utility of joining manipulations and trial functions together within a common space, leading them to mistake a localized virtual work manipulation upon a region of a trial function with a direct interaction between neighboring plates.[48] Viewed in linguistic terms, their employment of the word "element" secretly wobbles between "manipulation" and "trial function" readings depending upon applicational context. The upshot is a beautiful specimen of the phenomenon that I call semantic mimicry.

p. 357

(x)

The "impossible things" within my Physics 101 instruction stem from closely allied sources. One of the deepest difficulties of classical continuum mechanics (= the science of completely flexible bodies) was not fully clarified until the mid-twentieth century. I have in mind the problem of reconciling the two distinct kinds of "force" that emerge as relevant (this is a duality absent within the celestial mechanics circumstances considered earlier). If we inspect a small hunk of three-dimensional steel, we find two kinds of force at work: so-called body forces like gravitation that act across distances to pull directly upon local points p within our blob and traction forces that push and pull on the smallish surfaces that surround p. It is these traction forces that largely sustain the stress waves that ripple through our blob when we hit it with a hammer. A tricky reconciliation problem then arises because these two varieties of force have attached themselves to geometrically incompatible regions—the body forces to points and the traction forces to surfaces.[49] Even if we allow ourselves the pretense that we can look at a tiny region S that is so small that essentially only one body force acts inside S, we must still confront the fact that S's surrounding surface (conventionally designated as δS) must still carry an infinite bristle of traction vectors that need to be processed in some manner so that they can interact sensibly with the solitary body force at p.

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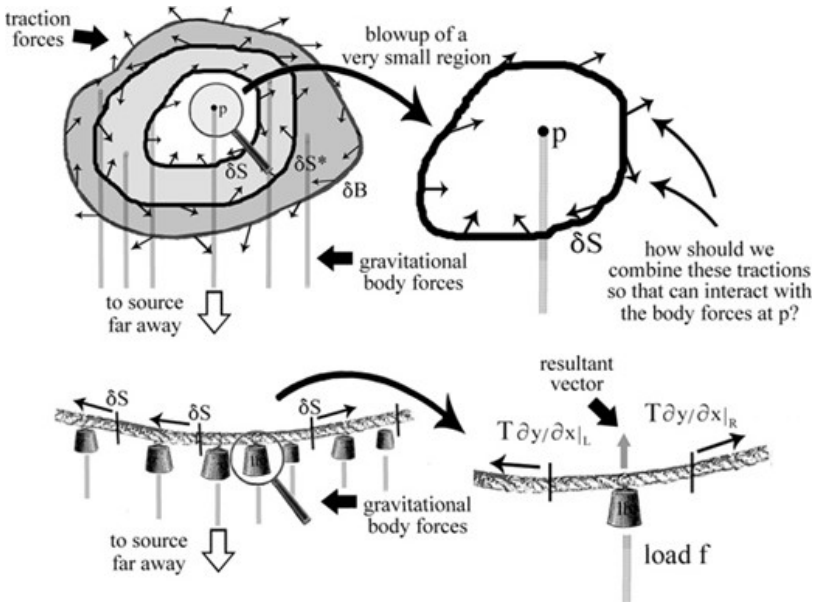

the traction/body force reconciliation problem

The modern resolution of this difficulty invokes the notion of a stress tensor living at p, a somewhat fancy mathematical construction that manufactures a p-centered force vector from the tractions to interact with the
p. 358 body forces after we slice through p with a ↳ selected plane (different slices produce different forces).[50] To obtain these unexpected characteristics, we need to shrink the little volumes S around p in the manner pioneered by Cauchy in the 1820s, although the official notion of a "tensor" wasn't properly articulated until the end of that same century. Sometimes, as the "global" and "local" appeals in the quotation from Oden and Reddy suggest, these shrinking processes do not result in a well-defined tensor, if, say, the point p lies on an interface between two regions of a different composition or a shock wave is passing through. One of the several schemes for implementing this "arrange matters globally before you look at points" policy is to invoke the integral formulations of virtual work policy cited above. From this point of view, Schwartz's "weak solution" manner of understanding the seemingly localized formula "$k\ d^2y/dx^2 = f(x)$" surrounds it with a flurry of test function variations that implement the requisite shrinking operations required to resolve our body/traction force reconciliation problem. In this way, many of the mysteries of Physics 101 procedure become rationally clarified, albeit at the cost of thinking about the mathematical structure of flexible bodies in a more sophisticated manner.

In the special circumstances of a one-dimensional string these underlying requirements become further hidden from view by the fact that the reconciliation of the traction forces at p can be adequately captured by a single vector, the "resultant" of the two endpoint tensions $T\partial y/\partial x|_L$ and $T\partial y/\partial x|_R$. But this is really a stress tensor hiding in vectorial clothing, and needs to be addressed as such. Some additional "impossible thing" folderol affects how these gizmos were manipulated back in Physics 101, but I'll not pursue those matters here.[51] However, the most striking irrationality of Physics 101 instruction lies in its claim that we can acceptably "idealize" a completely flexible string as a rigid body if we consider a small enough sector at equilibrium. From our analysis, we see that these appeals have merely served to dress an abstract technical issue—how to coordinate body forces with tractions?—in a sufficient amount of "familiar materials" garments that my instructor could stagger his way to the central models of classical continuum mechanics at the price of a few "impossible things," leaving only the unpleasant aftertaste of finding, in Hertz's words, that:

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stress tensor in vectorial clothing

p. 359

> [It is] *exceedingly difficult to expound to thoughtful hearers the very introduction to mechanics without being occasionally embarrassed, without feeling tempted now and again to apologize, without wishing to get as quickly as possible over the rudiments, and on to examples which speak for themselves.*[52]

Setting aside the jingoist colors in which he expressed such opinions, Pierre Duhem's celebrated opposition between the "imaginative minds of the English" and the "abstract minds of the French" can be retrospectively recast as a suspicion that the great modeling successes of Maxwell, Kelvin, Stokes, and others[53] were achieved via a semantic mimicry dressed in the misleading garments of "wires and bell-pulls." Pondering Kelvin in his pre-peerage guise as William Thomson, Duhem writes:

> *In order to explain the dispersion of colored light, W. Thomson considers material molecules to be veritable edifices with interacting rigid and elastic elements ... These are not the [abstract] materials with which Descartes sought to construct the "machine" of the world ... No. These are concrete objects, similar to those we see or touch: rigid or elastic solids, compressible or incompressible fluids ... W. Thomson affects to designate them with ordinary names. He calls them wires or bell-pulls ... The elements which make up matter are similar to those which we see around us every day. They may be fluid like water, compressible like air, elastic like steel, or flexible like a strand of silk. Their nature does not need to be defined philosophically. It suffices that their properties fall under the senses. The mechanisms they serve to make up are not destined to be grasped by reason; they are destined to be seen by the imagination.*[54]

Indeed, the functional analysis recalibration of the finite element method by applied mathematicians helps us understand why such dodgy modeling methods worked so well for such a long period of time. An extremely wide class of differential equation models can be ably approximated by simple choices of finite elements, which can then be arrayed in the familiar clothing of ordinary wires, bell-pulls, and square steel plates. Thus a linear triangular element becomes a pair of hinged rods; a quadratic element becomes a set of springs, and so forth. In their amazing Treatise on Natural Philosophy,[55] William Thomson (Lord Kelvin) and P. G. Tait built up many of
p. 360 the central continuum mechanics structures employed today, all obtained through ↳ use of the "imaginative" appeals of which Duhem complained. They reached these results by unwittingly operating within a finite element universe that neatly parallels a proper continuum mechanics development, in exactly the "search within a subspace" manner of the Rayleigh–Ritz principle discussed in section (ix). In this case, modern research has demoted the erstwhile "central conceptual structures" of continuum mechanics to a lower category of "somewhat misleading approximations" for the modern machinery of stress tensors, integral principles, constitutive equations, and all that.

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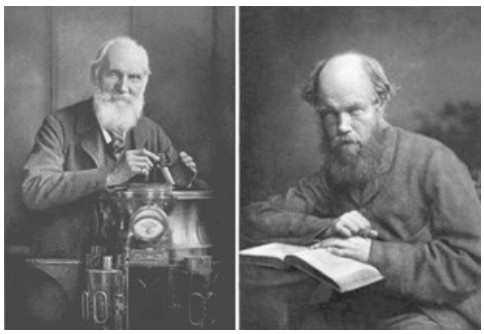

Thomson and Tait

Accordingly, when Duhem complains that these extraneous "imaginative pictures" provide little evidence for the existence of discrete molecular structures implementing these finite element renderings, his doubts have been largely ratified by the modern labors of the functional analysts.[56] We can add another worthy tenet to the catalog of Duhemian sentiments compiled in Essay 4.

(xi)

Three central themes emerge from our discussion that reverberate within our other essays. Descriptive tools that appear to capture natural circumstance aptly may not harmonize properly, as an inferential practice, with other descriptions that seem equally appropriate to their targets. Mathematicians have devised sophisticated repairs for these ineptitudes and, in doing so, have greatly enlarged the conceptual resources of mathematics. These constructions, which typically involve goodly amounts of set theory, were first devised by nineteenth-century giants such as Riemann and Dedekind, and later authors such as Schwartz and Sobelev have followed ably in their footsteps. On the relatively rare occasions when modern philosophers and mathematicians get together, the latter act publicly humble (but remain privately contemptuous)—"Oh, I'm not a philosopher at all; I only work on technical mathematical problems." If pressed, they will offer some crude version of formalism as the only "philosophy" to which they subscribe. These self-appraisals strike me as mistaken; significant amounts of substantial philosophizing, in any reasonable sense of the term, lie behind technical adjustments of the type surveyed in this essay. Indeed, the functional analysts have often served as subtler philosophers than most of us "professionals" in probing questions such as "How should the information encoded within a linear differential equation be properly viewed?" Replicating their successes in other walks of life requires a deep appreciation of underlying strategic rationale.

Secondly, adaptive development in language often hinges upon the indulgent cultivation of "impossible
p. 361 things." Looking over the long career of ever-improving classical ↳ mechanics, we realize that our modern understandings could have never been reached without the imaginative assistance of "infinitesimal element" window dressings; our modern functional analysis understanding of Physics 101 procedures could have never sprung fully wrought from Newton's brow. Looking back, we can only marvel, "Boy, we have finally assembled a fully rational account of how an effective continuum mechanics modeling should be pieced together, but our predecessors needed to bumble their way through a lot of strange methodological territory to get there." The pageant of unfolding conceptual adaptation contains dramatic shifts in applicational capacity reached through small reorientations of a fixed set of prior skills. Even to this day, we still tutor our pupils in "impossible thing" narratives, for such accounts recapitulate the adaptive phylogeny of the methods they must learn.

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advice from the Red Queen

Finally, the crucial turning points that open up fresh pathways of inferential exploration are often registered only contextually at first, in the guise of a tacit reliance upon the unacknowledged labor of "heuristic aids." The algorithmic assistance supplied by these seemingly out-of-territory sentences serves as a harbinger of unanticipated networks that implement novel forms of descriptive strategy.[57] Through such means, sophisticated forms of informational content remain encoded exclusively within the manner in which the usage contextually unfolds and may not be available to immediate inspection in the guise of stand-alone declarative sentences. When we first encounter the sophisticated inferential guidance of Schwartz's weak solution construction disguised in the masquerade of standard Physics 101 instruction, we may detect nothing beyond an uneasy sense of the methodological discomfort characterized by Paul Valéry:

> *You have surely noticed the curious fact that a certain* word, *which is perfectly clear when you hear or use it in* everyday *speech, and which presents no difficulty when caught up in the rapidity of an ordinary sentence, becomes mysteriously cumbersome, offers a strange resistance, defeats all efforts at definition, the moment you withdraw it from circulation for separate study and try to find its meaning after taking away its temporary function. It is almost comic to inquire about the exact meaning of a term that one uses constantly with complete satisfaction.*[58]

Notes

1 A. J. Davies, The Finite Element Method: A First Approach (Oxford: Oxford University Press, 1980), p. 1.

2 W. V. O. Quine, Methods of Logic (New York: Holt, Rinehart, and Winston, 1964); Israel Scheffler, The Anatomy of Inquiry (New York: Knopf, 1963).

3 This version parallels a famous remark of John von Neumann's about how one learns mathematics.

4 Stuart S. Antman, Nonlinear Problems of Elasticity (Berlin: Springer, 2005), pp. 11–12.

5 Here's a typical modern expression of such a Freezing Principle (from L. P. Lebedev and M. J. Cloud, Approximating Perfection (Princeton: Princeton University Press, 2004), p. 77):

> *The non-rigidity of the rope does not prevent us from using the classical mechanics of rigid bodies. When the rope is immovable in equilibrium, we cannot distinguish whether it is flexible or rigid. Thus, for a moment, we suppose it to be rigid and apply the laws of classical mechanics. In continuum mechanics, we call this the* solidification principle: *in equilibrium, any part of a deformable body can be considered as a rigid body under the reaction forces due to the remainder of the body, and so we can subject it to the laws of classical mechanics. Applying this to the rope, we find that the tension remains constant in each portion on which external forces do not act*.

For a good survey, see James Casey, "The Principle of Rigidification," Archive for History of Exact Sciences **43**(4) (1992),

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329–83.

6 For a survey, see my "Frege's Mathematical Setting" in Michael Potter and Tom Ricketts, eds, Cambridge Companion to Frege (Cambridge: Cambridge University Press, 2011).

7 Here I employ "harmonize" in the sense of Essay 2: the numbers credited to the boundary do not allow us to fill in the interior in a coherent manner.

8 J. N. Reddy, Theory and Analysis of Elastic Plates (Boca Raton: CRC Press, 1999), p. 73.

9 G. Barton, Elements of Green's Functions and Propagation (Oxford: Oxford University Press, 1989), pp. 158–9.

10 That is, we compute the integral $\int g_\alpha(x, y)\, h(x, y)\, d\alpha$.

11 In the circumstances Barton discusses (square boundaries), the resulting formula will display infinite stresses in its four corners but will otherwise supply quite accurate physical results elsewhere.

12 L. P. Lebedev, I. I. Vorovich, and Michael J. Cloud, Functional Analysis in Mechanics (New York: Springer, 2013), pp. 4–5.

13 Descriptive infirmities connected to the notion of "phase" have led modern researchers to articulate a sophisticated analog to the notion of "fundamental solution" within optics called a "wavefront set" based upon asymptotic Fourier series rather than simple impulses, as studied here (cf. Andrej Bóna and Michael A. Slawinski, Wavefronts and Rays as Characteristics and Asymptotics (Singapore: World Scientific, 2011)). These circumstances significantly underscore the morals advanced here, which I hope to detail in a future essay.

14 Essay 1 calls the increased computational demands of indeterminate bridges as the loss of an easy computational opportunity.

15 Due to the mechanical analogies with which they operated, they often tesselated planes into squares, whereas it is more common to employ triangles today.

16 J. T. Oden, Finite Elements of Nonlinear Continua (New York: McGraw-Hill, 1972), p. 4.

17 Here I refer to the rigidification doctrines referenced in footnote 4.

18 J. T. Oden, "Historical Comments on Finite Elements," in Stephen G. Nash, ed., A History of Scientific Computing (New York: ACM Press, 1996), p. 158.

19 We might call this return to centrality "Galileo's revenge," recalling his celebrated assertion (cited in Essay 6) that "the universe … is written in the language of mathematics, and its characters are triangles, circles, and other geometrical figures." Their triumphant return to physical relevance through their tesselational capacities illustrates one of my central themes—the difficulties of establishing a satisfactory accord between the representational tools that mathematics makes available to us and the uncooperative natural facts we must cover employing these mismatched representational ingredients.

20 Other descriptive ingredients such as constraints and modeling parameter ranges demand cooperative harmonies of their own, particularly in multiscalar circumstances. See Essays 1 and 5 for more on these behaviors.

21 Ernest Nagel, The Structure of Science (New York: Harcourt, Brace, and World, 1961), p. 32. In other books, we find to this same purpose: "antecedent conditions" (Carl Hempel); and "initial conditions" (Karl Popper). Scheffler employs all three phrases synonymously.

22 Or, more accurately, on a so-called Cauchy surface if no privileged temporal section is evident. In the case of delay equations, the term is often extended to cover data found within the backward characteristic cones of the current time slice.

23 But not always "lower dimensional," for sometimes one wants to tolerate three-dimensional "slushy regions" between areas governed by normal interior equations. In such cases, the cruder description pertinent to the slushy region takes the form of an interpolation rule which doesn't pretend to great descriptive accuracy with respect to the rather complex physical events that transpire in the transitional region.

24 We'll touch on such matters only fleetingly here, but the modern notion of a "trace deposit" indicates how delicate the required harmonizations between interior equations and boundary conditions can be.

25 David Armstrong, What is a Law of Nature? (Cambridge: Cambridge University Press, 1985), pp. 6–7.

26 On the difficulties that open up when we move away from point mass mechanics, P. G. Tait is eloquent (Heat (London: MacMillan, 1895), pp. 9–10):

> *[A]ll who have even a slight acquaintance with the subject know that the laws of motion, and the law of gravitation, contain absolutely all of* Physical Astronomy, *in the sense in which that term is commonly employed:–viz., the investigation of the motions and mutual perturbations of a number of masses (usually treated as mere points, or at least as rigid bodies) forming any system whatever of sun, planets, and satellites. But, as soon as physical science points out that we* must *take account of the plasticity and elasticity of each mass of such a system, the amount of liquid on its surface, … [etc.], the simplicity of the data of the mathematical problem is gone; and physical astronomy, except in its grander outlines, becomes as much confused as any other branch of science*.

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> For more on these issues, see my "Determinism: The Mystery of the Missing Physics," British Journal for the Philosophy of Science **60**(1) (2008), 173–93.

27 I should write "corrected forms of PDEs" because PDEs, strictly understood, usually encounter harmonization difficulties of the sort surveyed in this essay. As a result, PDE models often require various forms of functional analysis tweaking before they can model continuous matter plausibly.

28 I can't recall where I picked up this descriptive phrase but it's a good one.

29 Such policies remind me of the practice of indiscriminately stuffing all of those unattractive Christmas gifts into the attic. Every so often Aunt Katherine pays a visit and you'd better recover that ugly vase pronto.

30 Not only are the cross-dimensional operations of trace operators and all that harder to explain, it turns out that their harmonization issues represent a red herring with respect to our aircraft engineers' motivating circumstances, for the "square boundaries" they thought they were welding together turn out to not represent proper boundaries at all, but mere simulacra of such divisions. This substantial revision in motivational picture forms the chief subject of this essay.

31 Virtually all good histories of mathematics contain nicely detailed accounts of this clash. Some samples: A. P. Youschkevitch, "The Concept of Function up to the Middle of the Nineteenth Century," Archive of the History of the Exact Sciences **16**(1) (1976), 37–85 and Clifford Truesdell, "The Rational Mechanics of Flexible or Elastic Bodies" in Leonhard Euler, Opera Omnia, Series II, Vol. 12 (Lausanne: Orell Füssli, 1954).

32 The construction of "fundamental solutions" suitable for many evolutionary equations (= of hyperbolic signature) often rely upon a preliminary collapse (via separation of variables) onto a "base characteristic manifold" of elliptic type, such as are directly obeyed by systems in equilibrium. In this essay I will avoid the mathematical tricks required to lift type 1 Green's functions into type 3 Green's functions.

33 Note that that no strain energy gets stored within the two straight portions of the $V_\alpha(x)$ curve.

34 I shall follow the standard (if unjust) policy of speaking of Schwartz as the sole originator of these techniques, even though Sobolev wrote earlier and Schwartz knew of his work. In his Mathematical Understanding of Nature (Providence: American Mathematical Society, 2014), p. 123, V. I. Arnol'd credits an earlier author, N. M. Gunter, with many of the central ideas.

35 Jean Dieudonné, quoted in Jesper Lützen, The Prehistory of the Theory of Distributions (Berlin: Springer, 1982), p. 3.

36 The greatest crisis of this sort centered upon Riemann's appeals to Dirichlet's principle within complex function theory.

37 J. T. Oden and J. N. Reddy, Variational Methods in Theoretical Physics (Berlin: Springer, 1983), p. 13.

38 F. G. Friedlander, Sound Pulses (Cambridge: Cambridge University Press, 2009) p. 40.

39 Robert S. Strichartz, A Guide to Distribution Theory and Fourier Transforms (Singapore: World Scientific, 2003), p. 2.

40 Strictly speaking, we move into an arena in which our official modeling tools become equivalence classes of functionals, within which one can sometimes locate old-fashioned functions as "representatives." Of this policy, Dorothee D. Haroske and Hans Triebel remark:

> *This is reminiscent of the famous final slogan in G. Orwell's novel Animal Farm, which reads, adapted to our situation: "All representatives of an equivalence class are equal, but some representatives are more equal than others."* (Distributions, Sobolev Spaces, Elliptic Equations (Zurich: EMS Publishing, 2007), p. 155)

41 These extended tolerances are commonly called "weak solutions," to distinguish them from the old-fashioned "classical solutions" explicated by Cauchy and Weierstrass.

42 Similar ideas can be applied to the trickier problem of bringing the class of expected boundary conditions into improved harmony with our interior modelings. Characterizing a sheet of metal as a perfect square seems no more "idealized," from an observational point of view, than assigning it tightly rounded corners. But earlier we were forced to reject boundary condition characterizations of a perfect square ilk because our interior equations didn't like 'em. But privileging the interior's demands in this one-sided manner seems unjust (and linguistically inconvenient). In advanced textbook treatments, "possible boundaries" and "possible interiors" are located in separate spaces of differing dimensions and Schwartz-style mollifers are employed to allow the latter to deposit distribution-like "traces" upon the former. I won't discuss further details here, because they are somewhat orthogonal to our central argument.

43 Typically $\delta y(x)$ can be any smooth function satisfying the problem's boundary conditions. Note that here $\delta y(x)$ is a variation, not a Dirac δ-function.

44 Observe that one of the derivative operations ($\partial/\partial x$) formerly applied to $y(x)$ has been transferred to the variation $\delta y(x)$ by a tacit integration by parts manipulation. The resulting formula, however, has wider applicability than the point-centered differential equation from which it derives. The Lagrangian point of view advocated by Oden and Reddy claims that mechanics should be initially formulated in terms of these integral formulations (or some variant thereof), rather than founded within a purely localized **F** =m**a** starting frame. These methodological precepts are now accepted as orthodoxy

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within modern continuum mechanics. We'll find that the "impossible things" of my Physics 101 instruction trace to a failure to approach the content of our string equation in this rather complex manner.

45 Oden and Reddy, Variational Methods in Theoretical Physics, pp. 1–2.

46 The same logic allows us to consider $\mathfrak{S}_{\mathrm{FE}}$ comprised of simple curved approximates such as quadradics. Sometimes such decompositions produce more accurate results more quickly.

47 William Gilbert Strang and George J. Fix, An Analysis of the Finite Element Method (Wellesley: Wellesley-Cambridge Press, 1973), p. 2. This book is one of the early classics of functional analysis diagnosis.

48 Considered closely, we find that the hat function elements employed overlap one another and that no side-by-side computations of a true boundary interface character truly appear, contrary to the original impressions of our aircraft engineers.

49 More accurately, each represents a density that requires a volume or surface integral to supply a finite force.

50 I apologize for the brisk opacity of this description. Any modern text on continuum mechanics explains the construction at greater length.

51 I particularly have in mind the assumption, marked as "impossible thing" (2) above, that the resultant vector points entirely in a vertical direction. As nicely explained in Stuart S. Antman, "The Equations for Large Vibrations of Strings," American Mathematical Monthly **87**(5) (1980), 359–70, this is the consequence of substituting a geometrical constraint for a constitutive assumption with respect to stress. Characteristic conceptual puzzles that arise from the invocation of constraints are discussed in the appendix to Essay 7.

52 Heinrich Hertz, The Principles of Mechanics, trans. D. Jones (New York: Dover, 1956), pp. 6–7.

53 By any plausible standard, the results they reached are truly stupendous. The locus classicus for Duhem's obnoxiously rendered opinions is The Aim and Structure of Physical Theory (Princeton: Princeton University Press, 1954), pp. 55–104.

54 Pierre Duhem, "The English School and Physical Theories" in Essays in the History and Philosophy of Science, trans. Roger Ariew and Peter Baker (Indianapolis: Hackett, 1996), p. 57.

55 W. Thomson and P. G. Tait, Treatise on Natural Philosophy (New York: Dover, 1962) (retitled as Principles of Mechanics and Dynamics).

56 For a brief survey of these suspect arguments, see Wandering Significance, pp. 656–7. On Kelvin's behalf, we might note that a definitive articulation of the abstract balance principles we should trust in, while shrinking our little volumes down to points, remains controversial—how many allowances for point-based "directors" in the mode of Cosserat media should be tolerated, for example? On this, see M. B. Rubin, Cosserat Theories: Shells, Rod and Points (Dordrecht: Kluwer, 2000). In view of such uncertainties, verifying that we can actually construct a gizmo that transmits thrusts in the manner posited strikes me as a reasonable precaution.

57 In Wandering Significance, pp 518–22, I cite the sly recommendations of Oliver Heaviside on this score: "Shall I refuse my dinner because I do not fully understand the process of digestion?" (Electromagnetic Theory, Vol. II (New York: Chelsea, 1971), p. 9). I am as confirmed a "scientific realist" as walks this planet and believe that ongoing science gradually accumulates a large set of reliable "truths." However, I don't believe that the truth-rules for these truths can be captured by the simple isomorphisms favored by ontic structural realists and similar thinkers (see John Worrall, "Structural Realism: The Best of Both Worlds" in D. Papineau, ed., The Philosophy of Science (Oxford: Oxford University Press, 1996). Such views, in my opinion, suffer from insufficiently flexible opinions with respect to the semantics of language.

58 Paul Valéry, The Art of Poetry, trans. Denise Folliott (Princeton: Princeton University Press, 1958), p. 54.

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